5G E2E QoS
(Quality of Service)

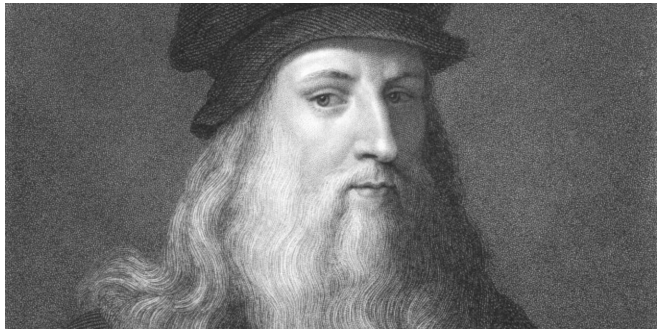
Created: 2024_0518

Updated: 2025_0518

By JMA

Org_Resource

翔宇科技
EAGLETEK



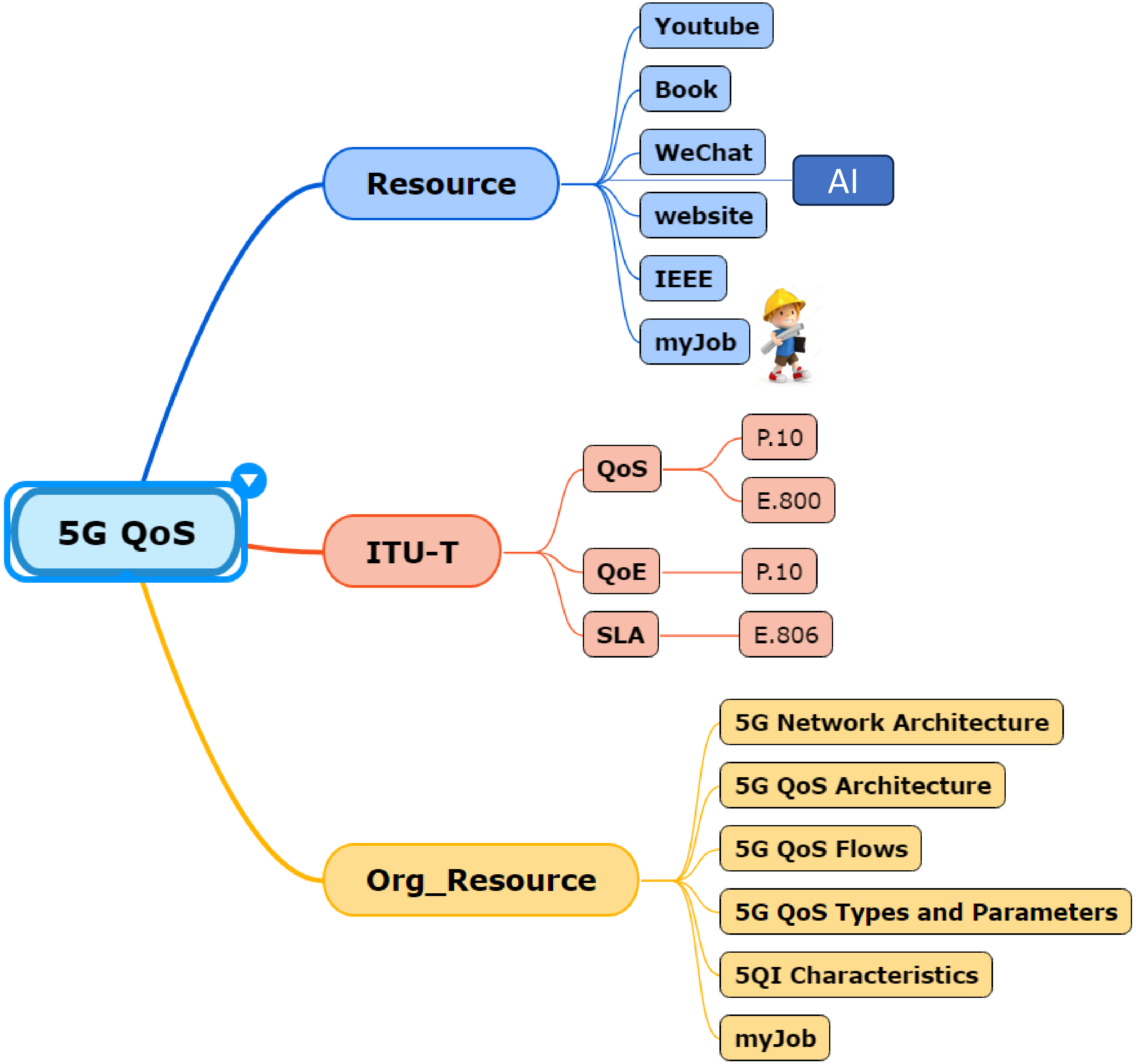
He who has access to the fountain does not go to the water-pot.

~ Leonardo da Vinci ~

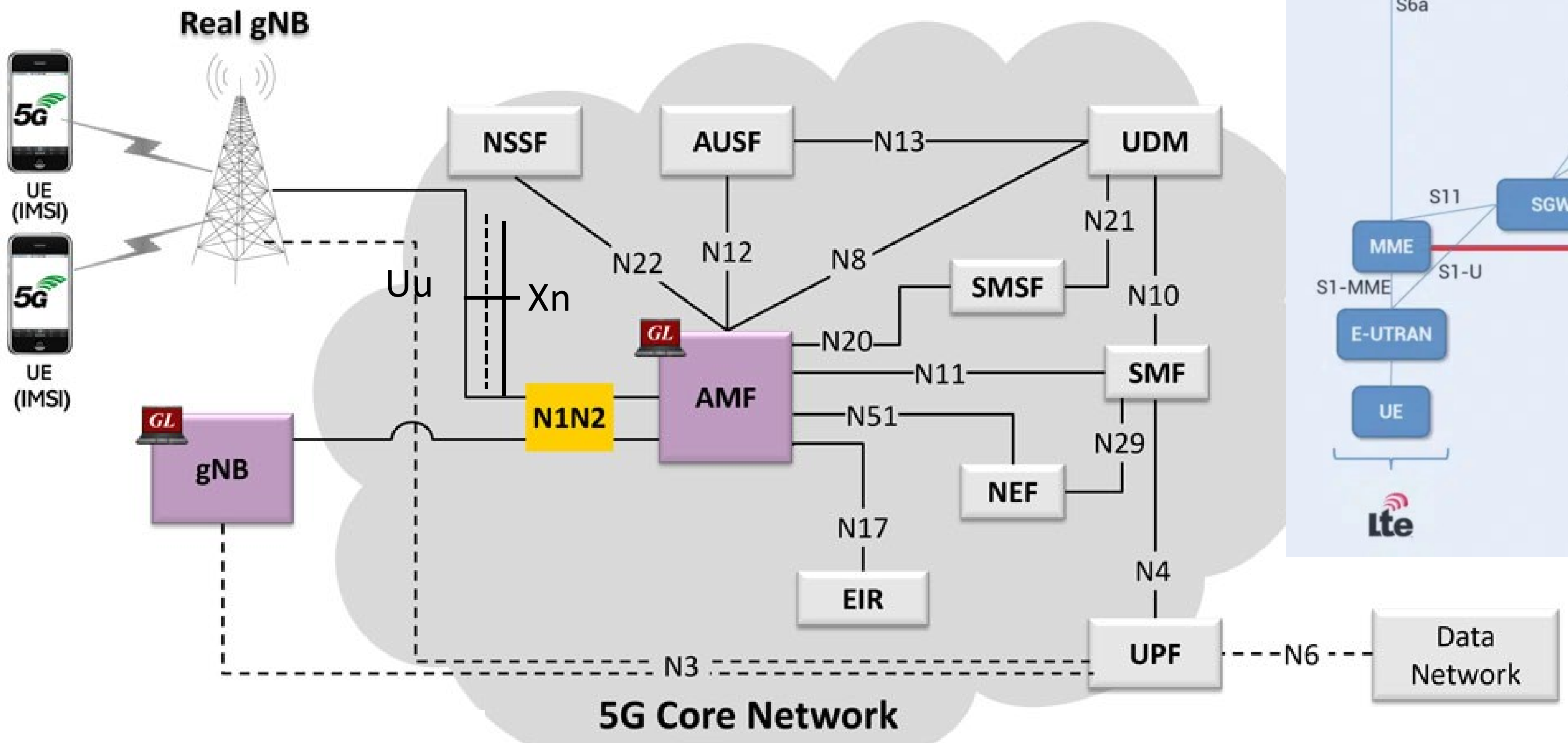
5G E2E QoS Call Flow

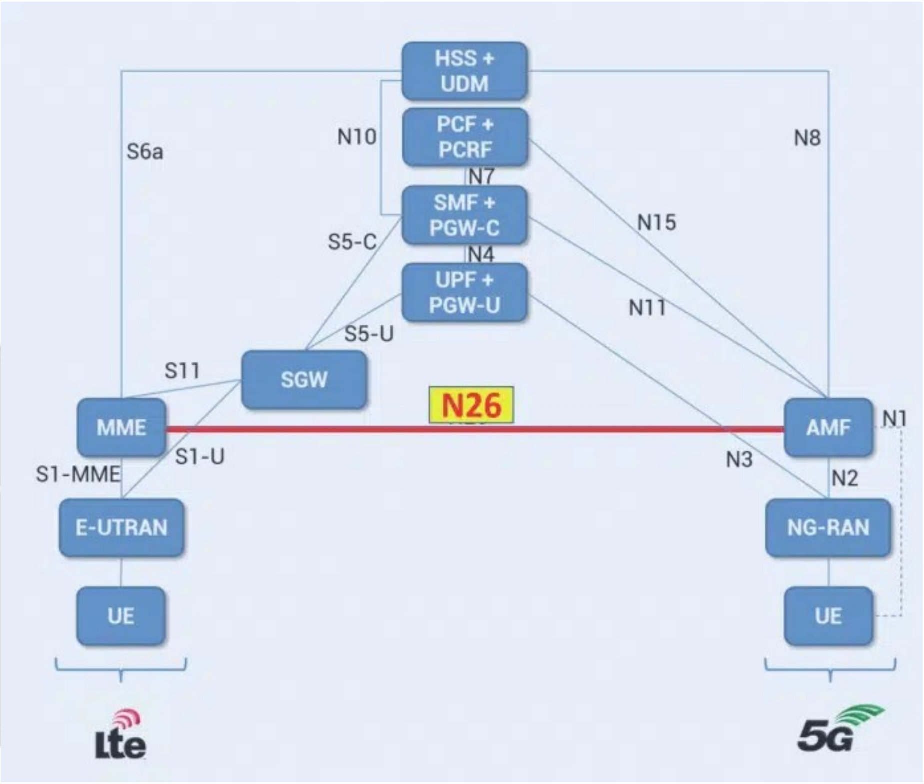
60min

翔宇科技
EAGLETEK


5G QoS
Resource
Youtube
Book
WeChat
website
IEEE
myJob
AI
ITU-T
QoS
P.10
E.800
QoE
P.10
SLA
E.806
Org_Resource
5G Network Architecture
5G QoS Architecture
5G QoS Flows
5G QoS Types and Parameters
5QI Characteristics
myJob

5G SA Network Architecture



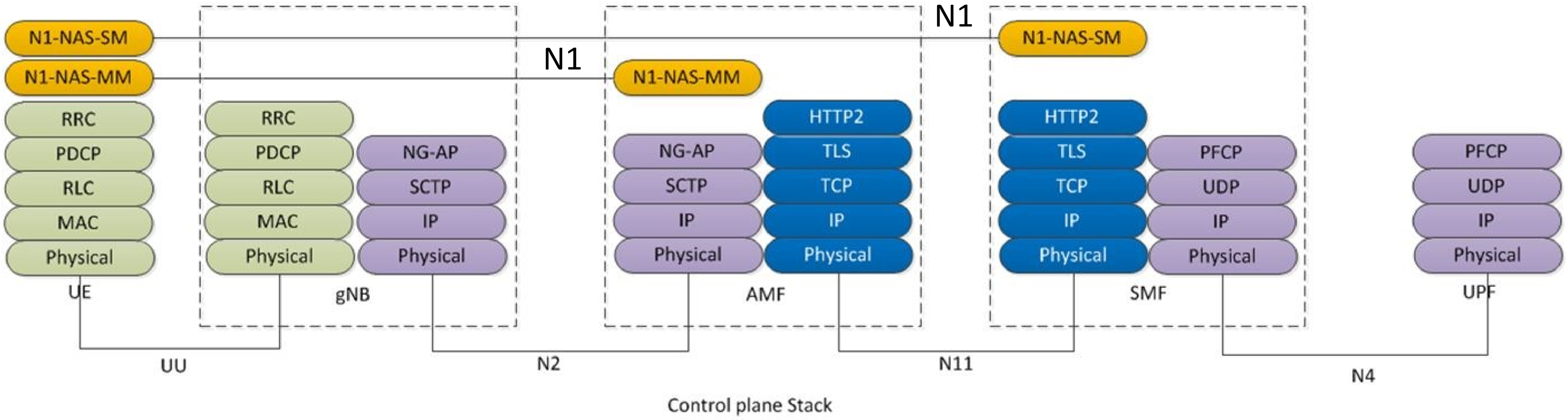


Source: https://www.5gworldpro.com/blog/2020/11/17/is-interface-n26-important-in-5g/

翔宇科技
EAGLETEK

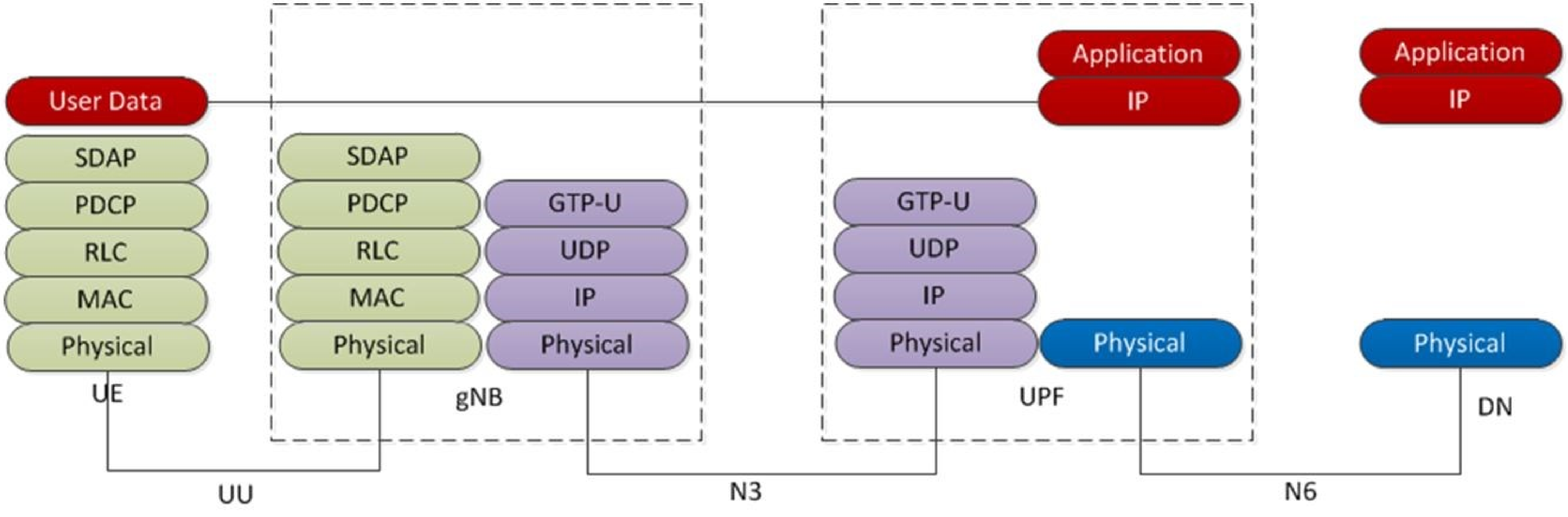
source: https://www.gl.com/5G-N1N2-interface-emulator-using-maps.html

5G E2E: Uu, N1, N2



Ref: www.5gblogs.com/5g-network-architecture/

5G E2E: Uu, N3



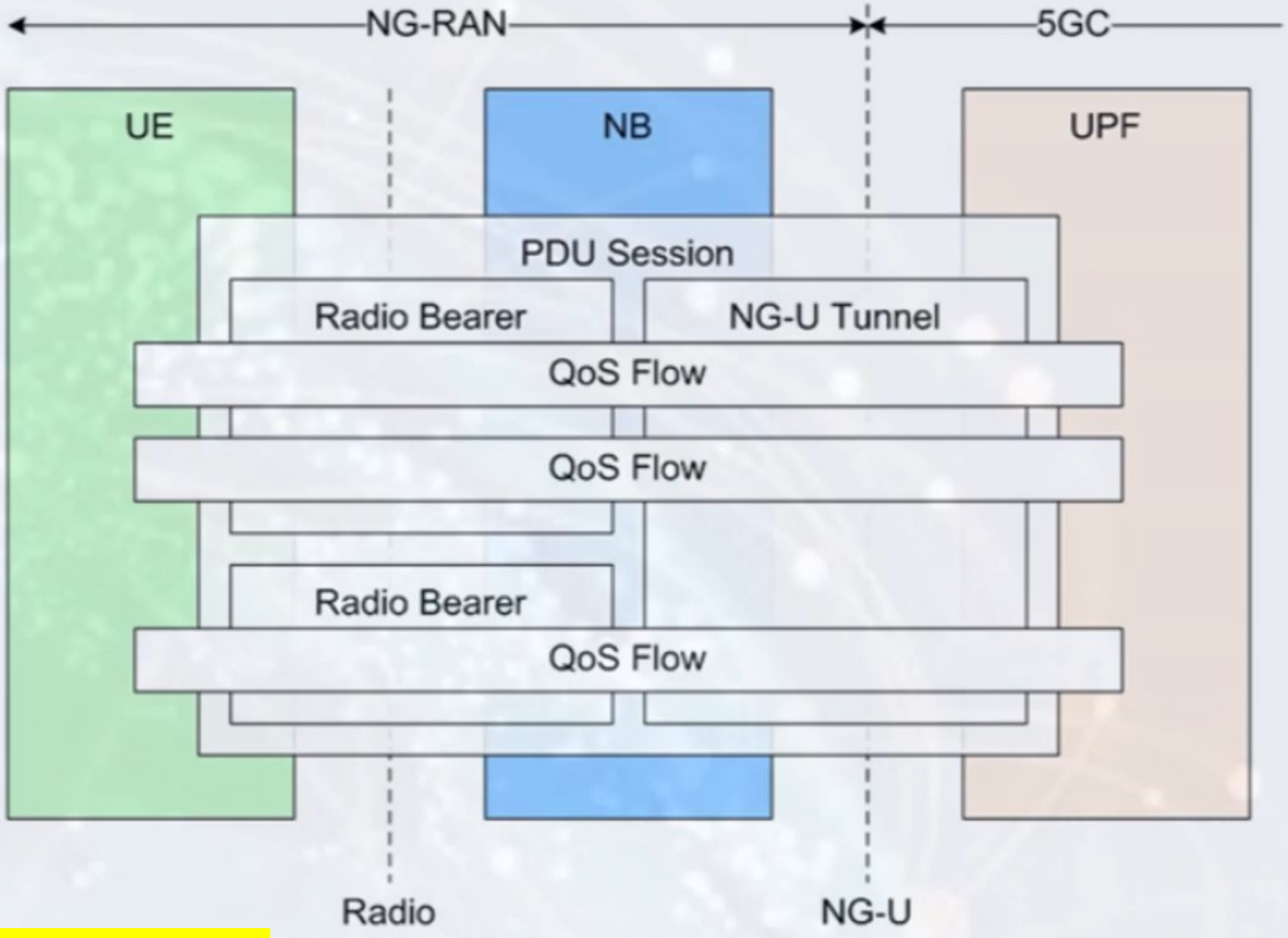
User plane Stack

Ref: www.5gblogs.com/5g-network-architecture/

5G QoS Architecture

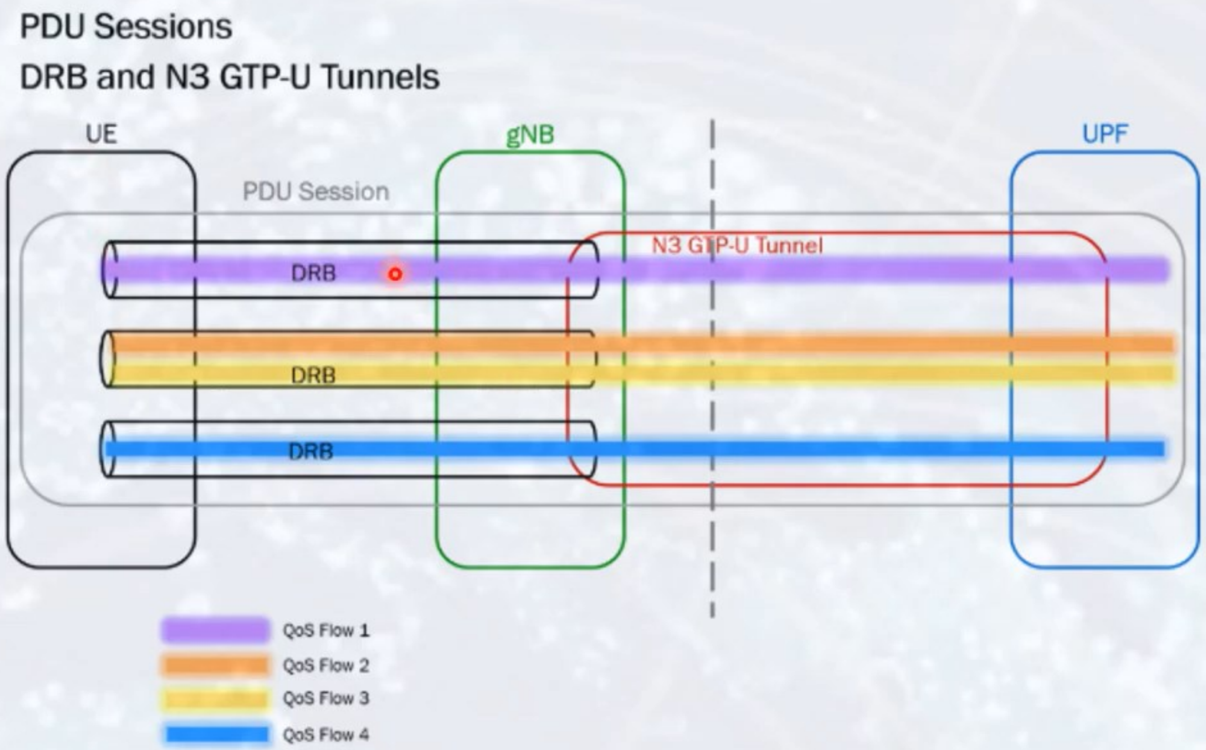
Types of QoS Flow

- QoS Flow with GBR
- QoS Flow with NGBR
- Delay Critical GBR
- Reflective QoS



https://youtu.be/KS4bOW3J8m4?si=tl5Res-uFehBGf9v

5G QoS Architecture



A PDU session may contain multiple QoS flows and several DRBs but only a single N3 GTP-U tunnel. A DRB may transport one or more QoS flows.

Data Radio Bearer

~~DRB : Dedicated Radio Bearer~~
GTP : GPRS Tunnel Protocol

https://youtu.be/KS4bOW3J8m4?si=tl5Res-uFehBGf9v

5G QoS Architecture

https://clcnetwork.wordpress.com/2020/04/19/5g-core-pdu-session-and-qos-part-2/

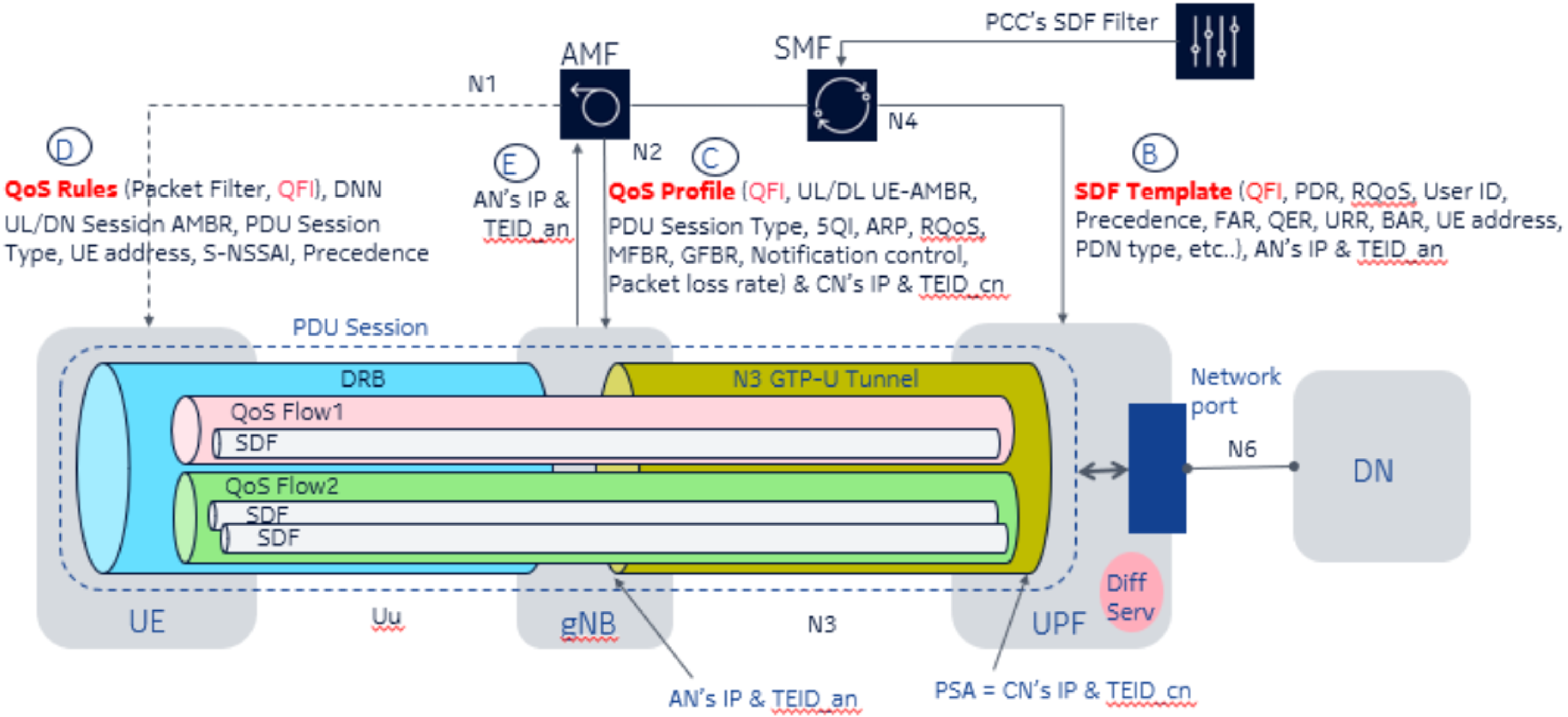


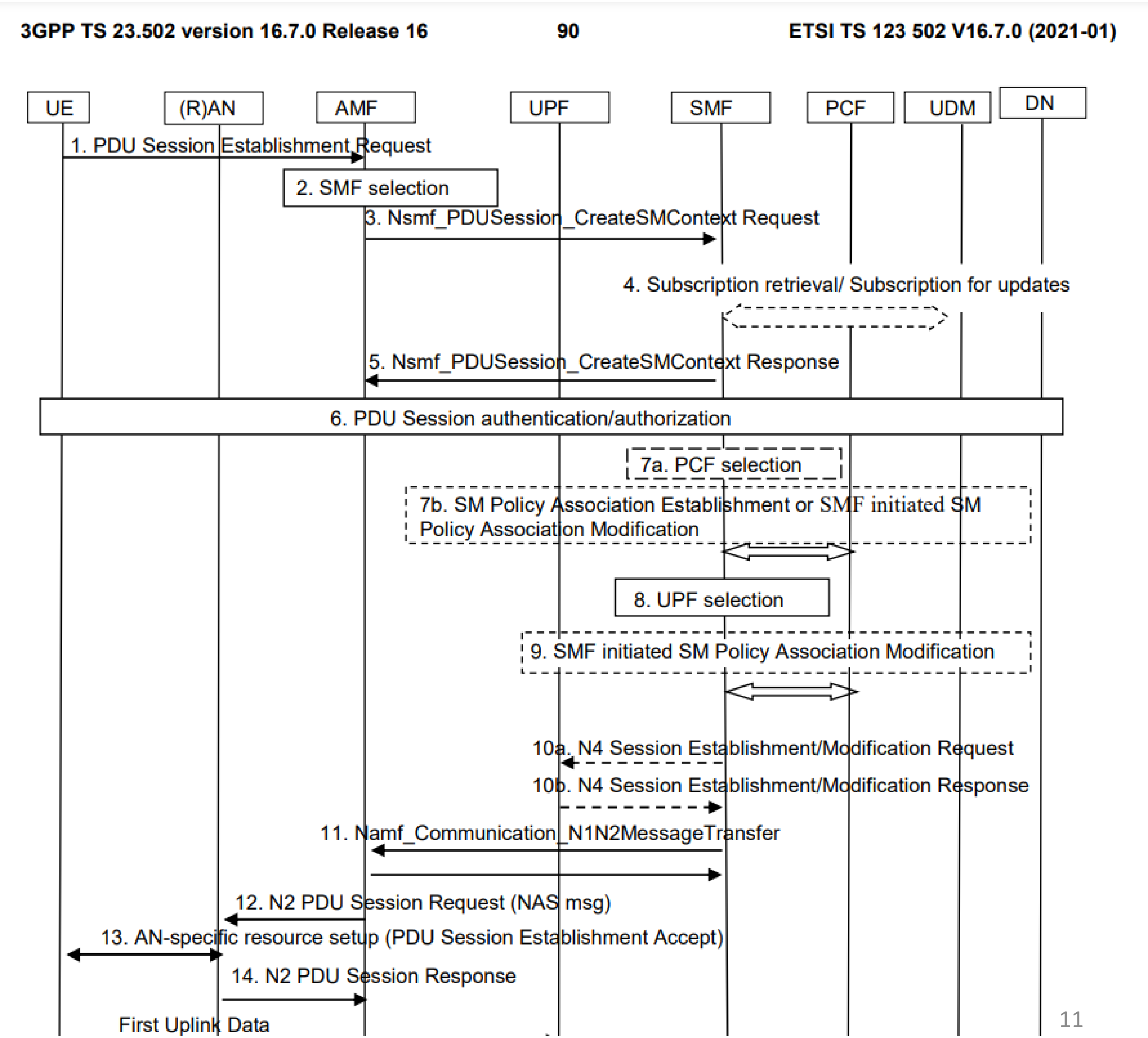
Figure 1 – SDF Template, QoS Profile and QoS Rule for UPF, gNB and UE respectively

5G QoS Call Procedures

https://clcnetwork.wordpress.com/2020/04/19/5g-core-pdu-session-and-qos-part-2/

- In order for a PDU session and a default QoS Flow to be established along the UE, gNB to UPF to carry SDFs (Service Data Flows) or traffic, a lot of IEs (Information Elements) are exchanged among AMF, SMF, UE, gNB and UPF as shown in the Figure 1:
- The SMF directly and indirectly (via the AMF and the gNB) sends the below QoS constructs for the PDU Session and the default QoS Flow Establishment:
  - SDF Template to UPF
  - QoS Profile to gNB
  - QoS Rule to UE
- Beside setting up the PDU Session and the default QoS Flow comprising a non-GBR Data Radio Bearer (DRB) between the UE and the gNB and the N3 GTP-U tunnel between gNB and UPF, the above IEs in the Figure 1 are also used for QoS processing and enforcement of the UE's traffic such as:
  - Classification — Classify or detect the UE's uplink (UL) and downlink (DL) traffic onto the appropriate QoS Flows within the UE's PDU session
  - Queuing and Scheduling — Enforce GBR and non-GBR UE traffic such as UL/DL bandwidth, latency, traffic priority etc…
  - Marking/Remarking — Mark the mobile traffic when it leaves the 5GC to the DN so that the priority and QOS of the traffic can be honored in the DN

5G QoS Call Procedures



3GPP TS 23.502 version 16.7.0 Release 16

5G QoS Call Procedures

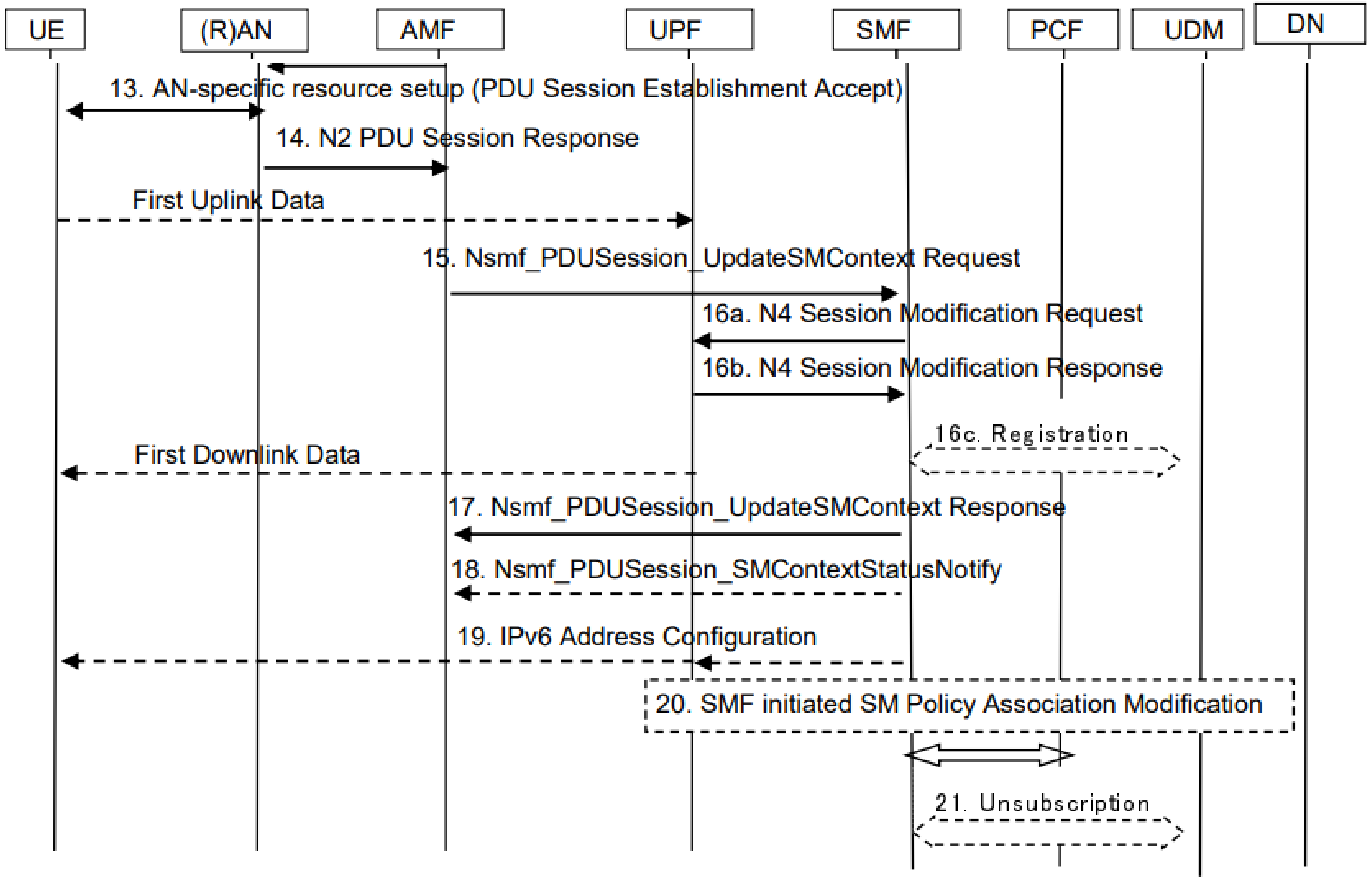


Figure 4.3.2.2.1-1: UE-requested PDU Session Establishment for non-roaming and roaming with local breakout

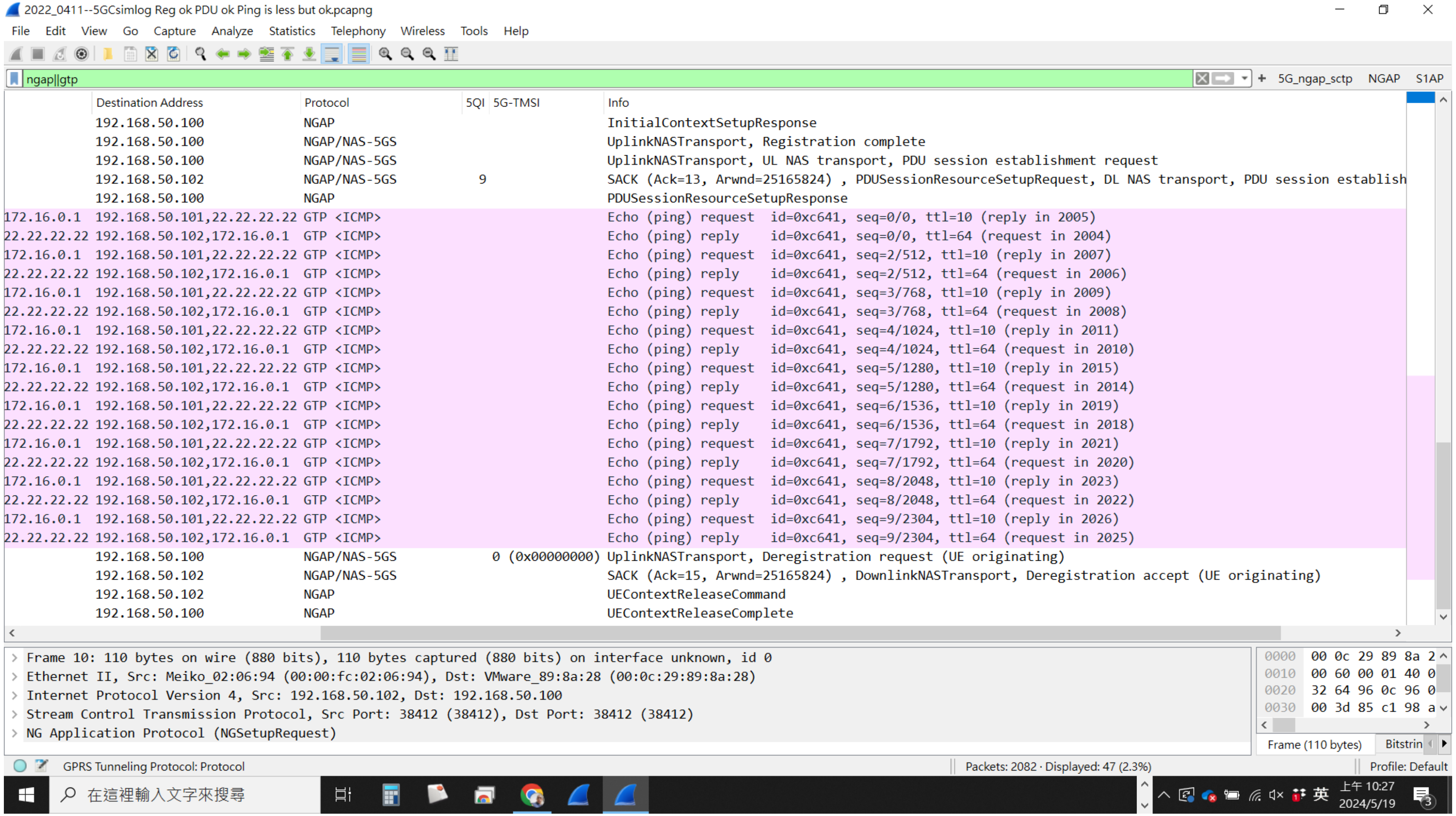
3GPP TS 23.502 version 16.7.0 Release 16

5G QoS Callflow in 5GCsim

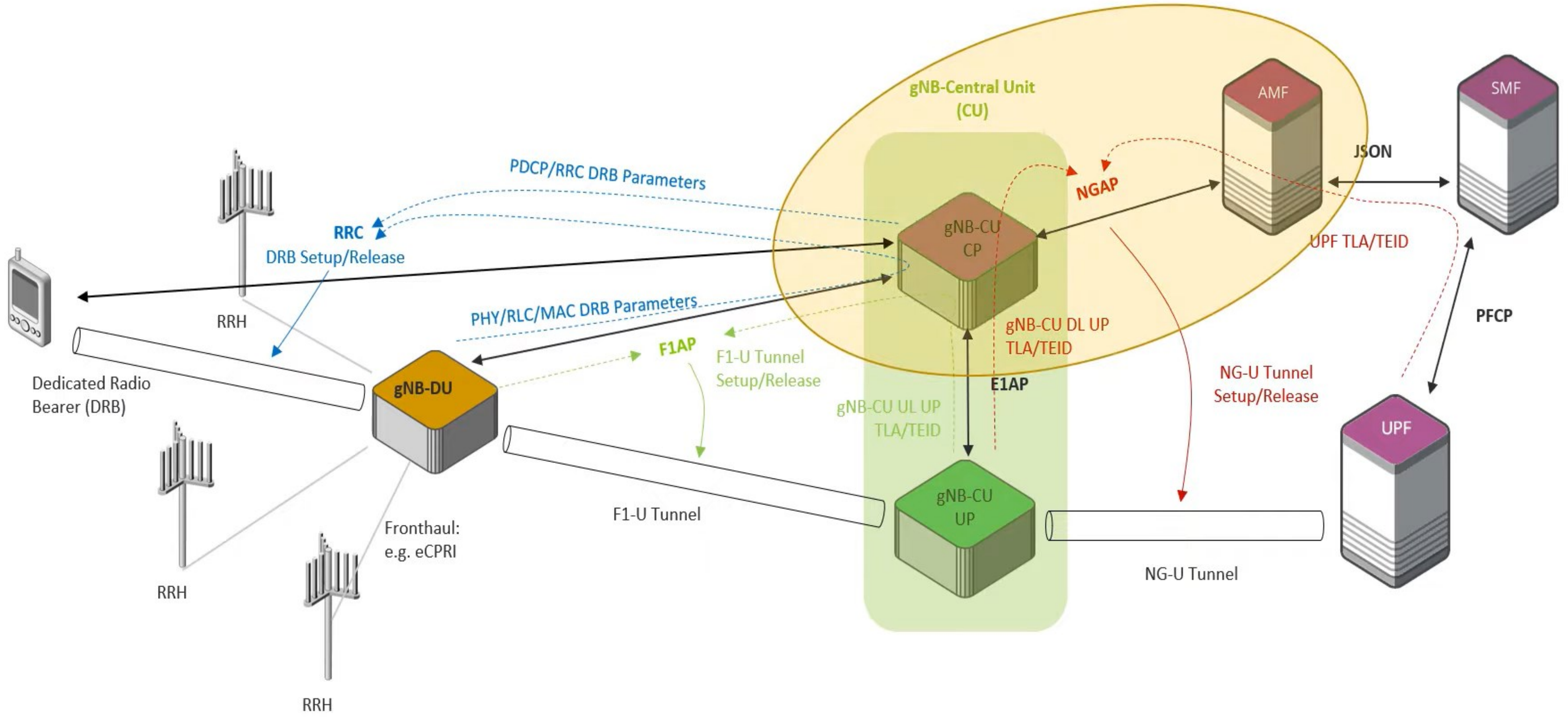
2022_0411--5GCsimlog Reg ok PDU ok Ping is less but ok.pcap

- 5G SA
- 1 PDUsession
- With default 5QI=9, ping 10s
- Save E2E logs

5G QoS Callflow – after Registration

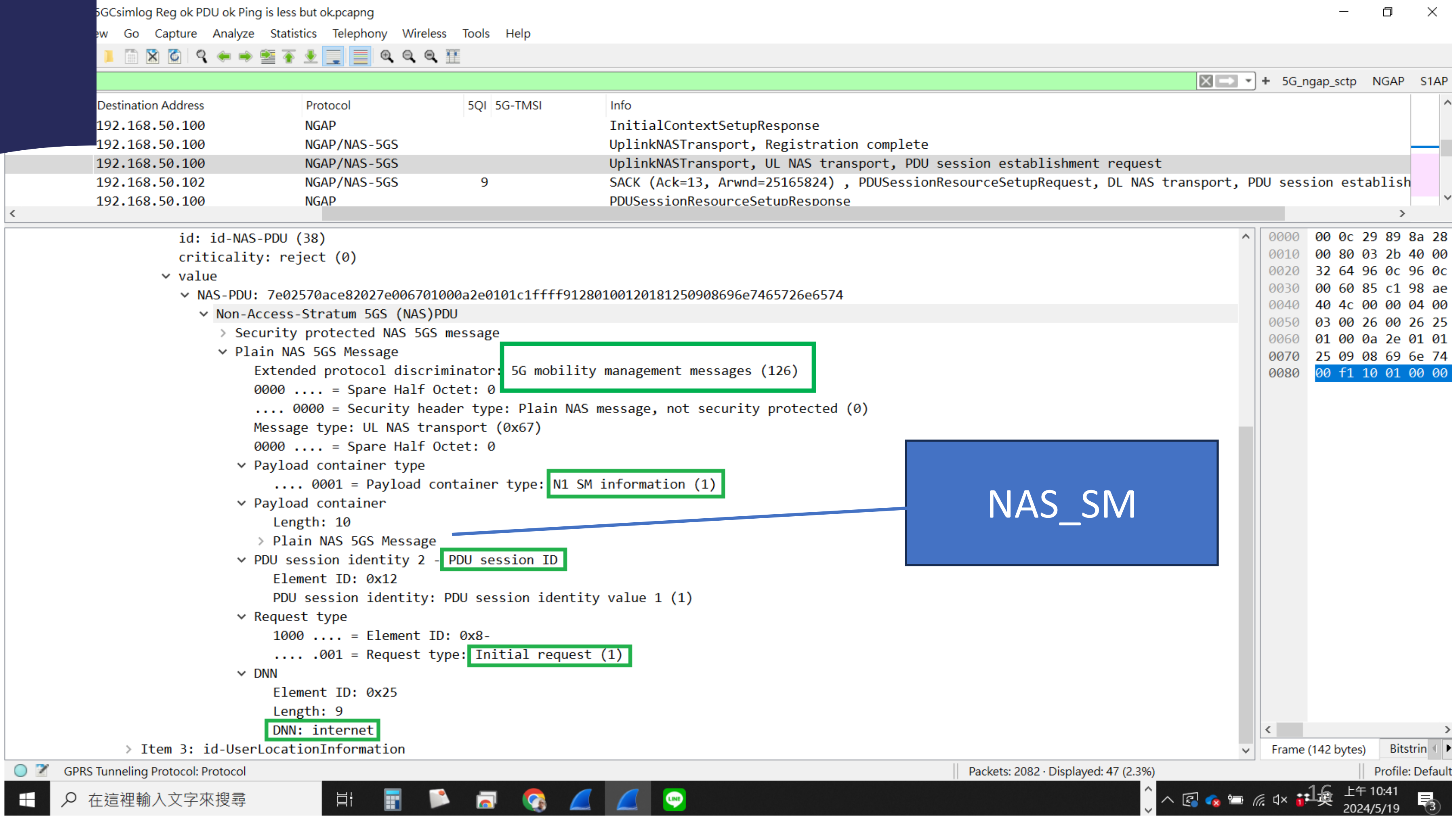


Protocols Involved in Establishing UP Transport

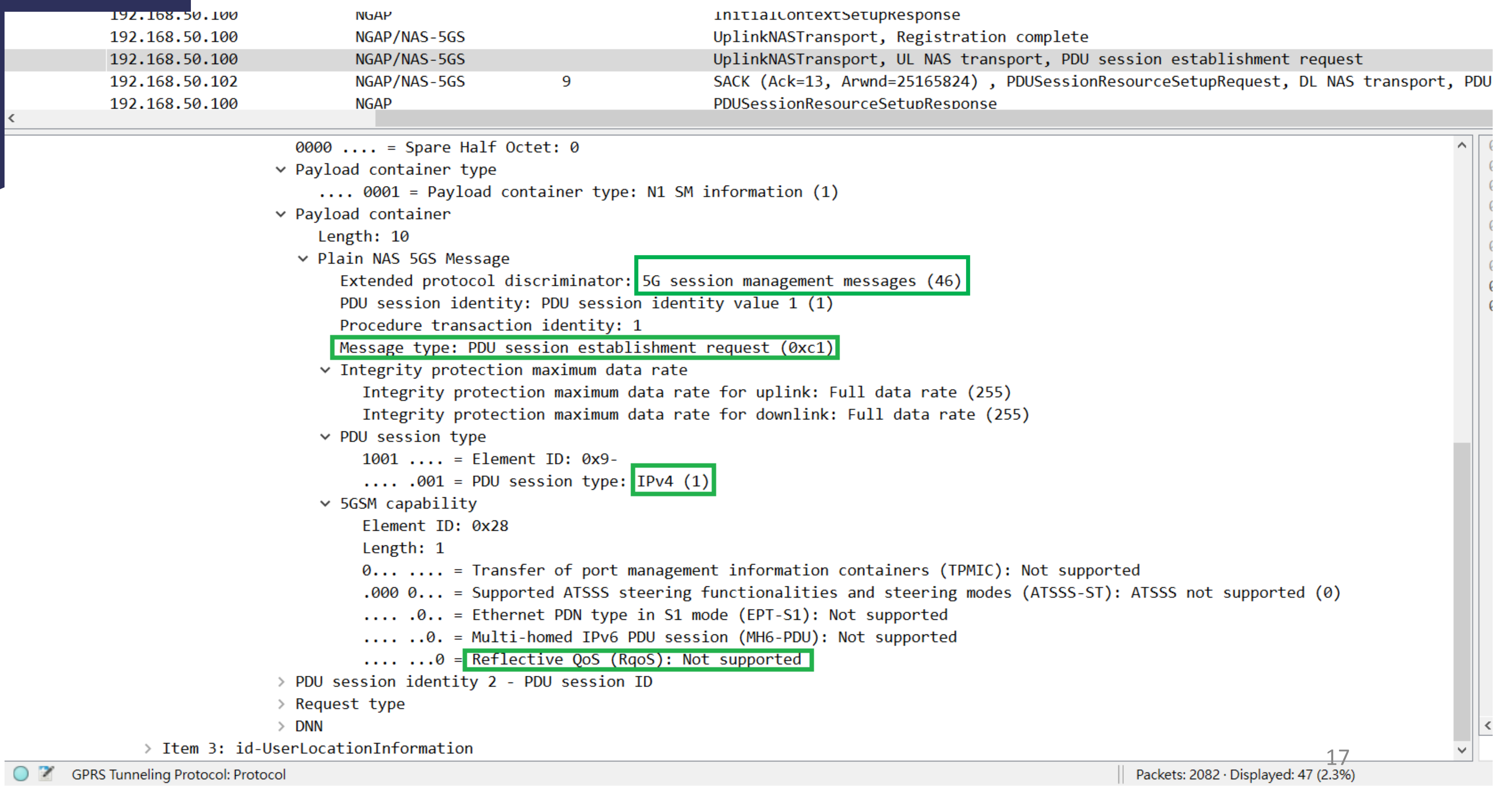


https://youtu.be/fnii_fA1W4s?si=lJIrD1JGTXMmE8mk

5G QoS Callflow – PDUsessionEst /NAS_MM



5G QoS Callflow – PDUsessionEst /NAS_SM

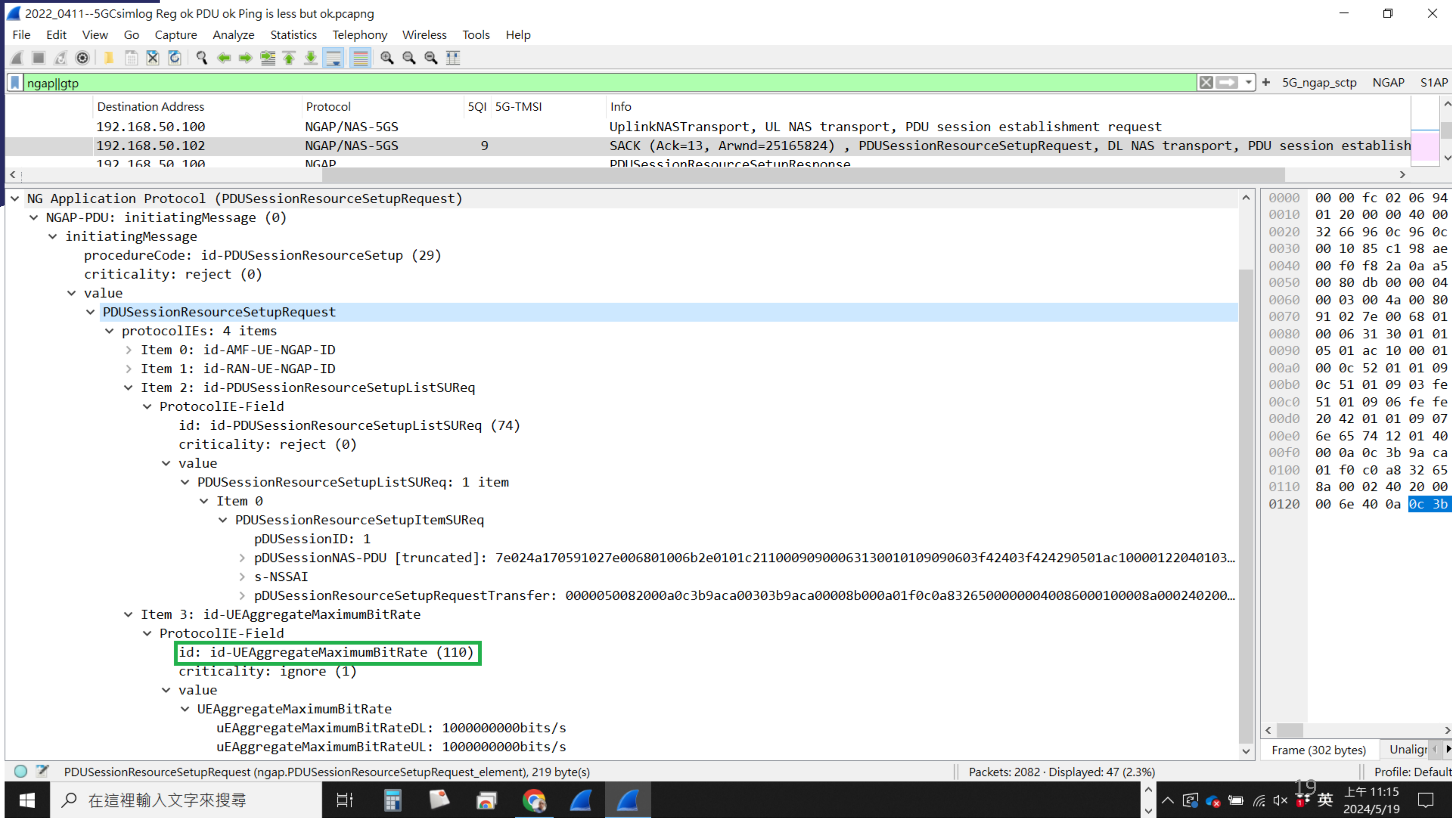


5G QoS Callflow – translations for humen--1

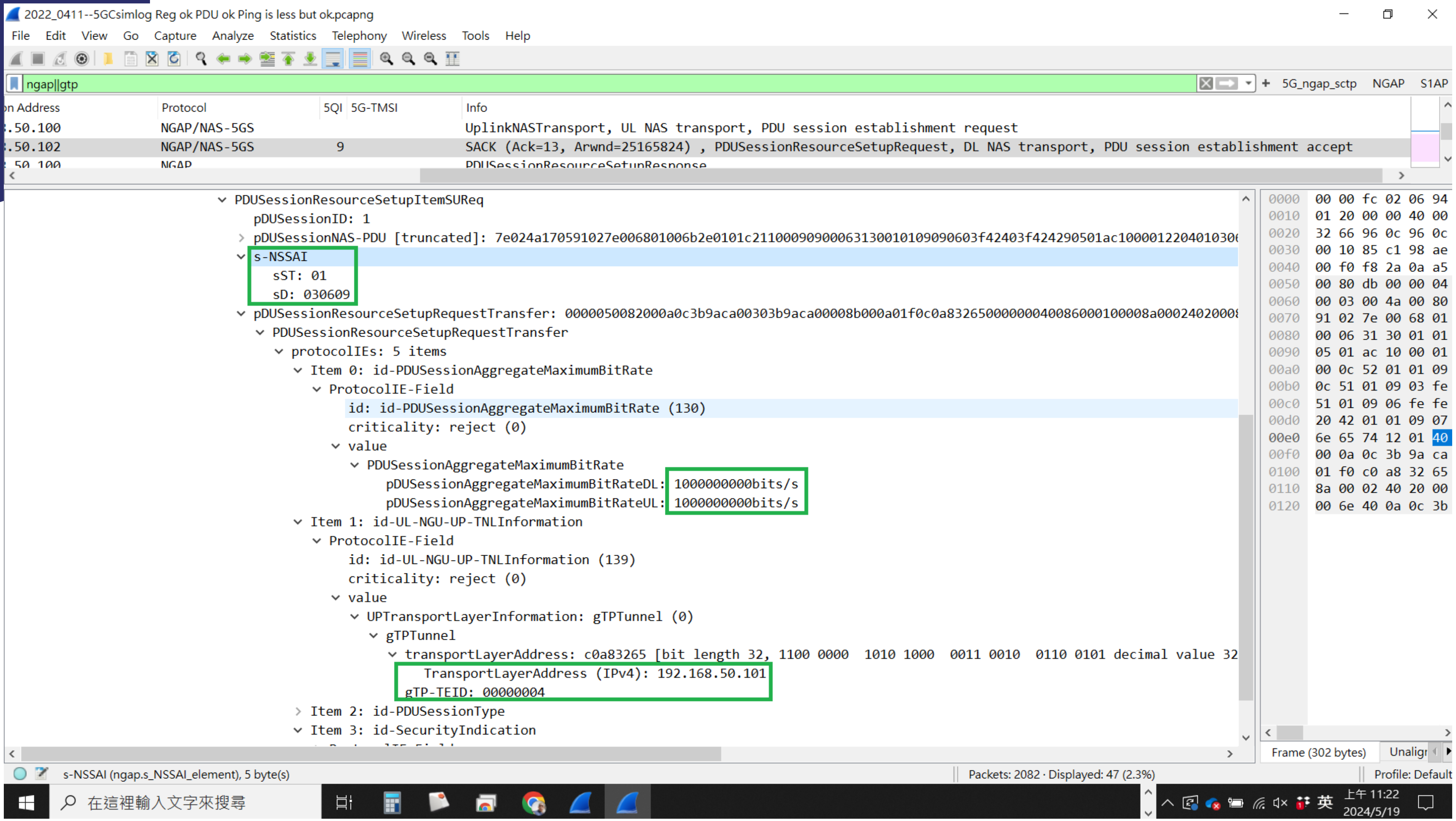
UE to AMF:

- It is initial request to AMF
- To which DNN
- With which PDUsession ID
- PDUsessionEstRequest
- Max Data rate
- IPv4 or IPv6
- 5G SM Cap

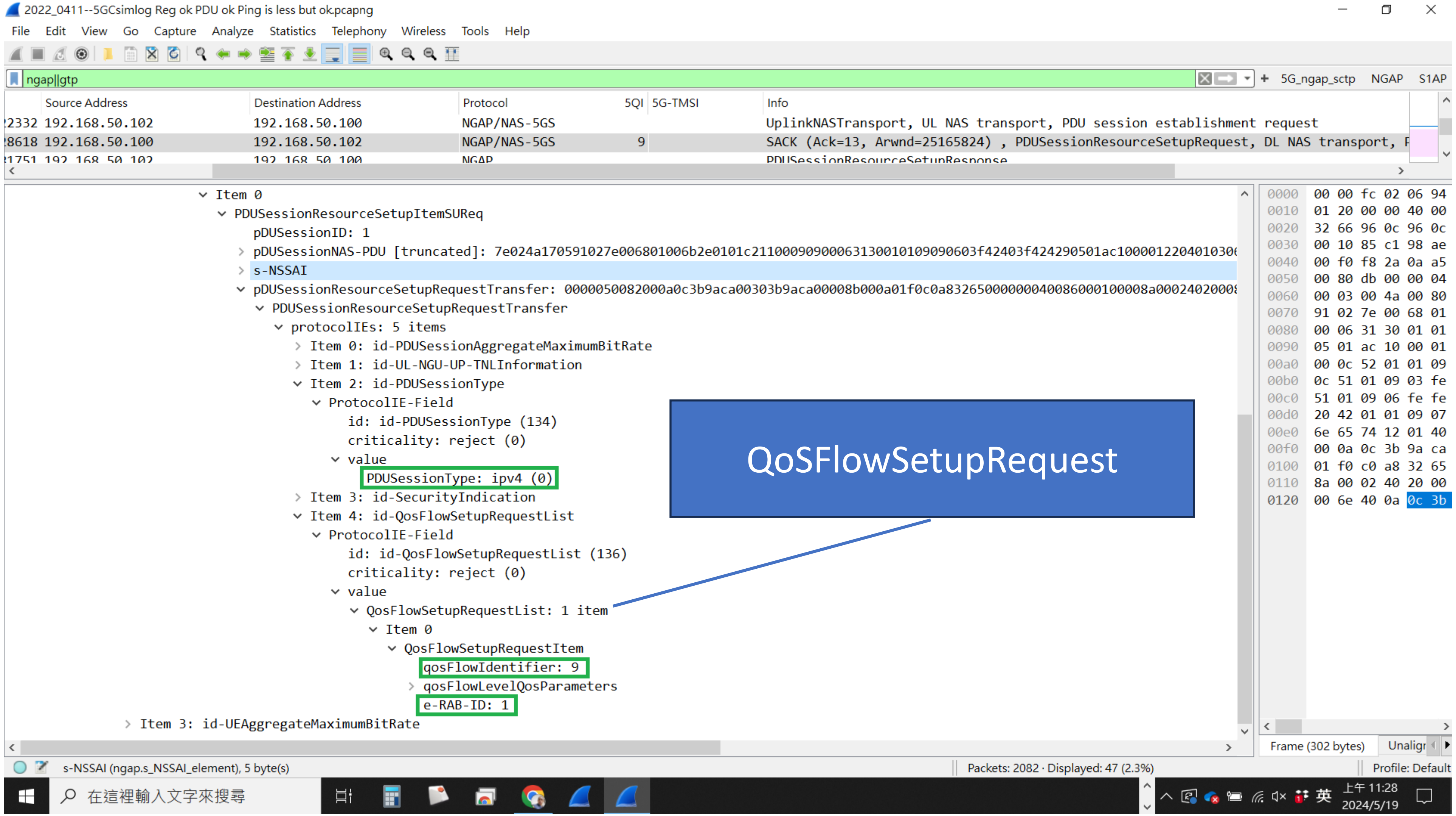
5G QoS Callflow – PDUsession Resource --1



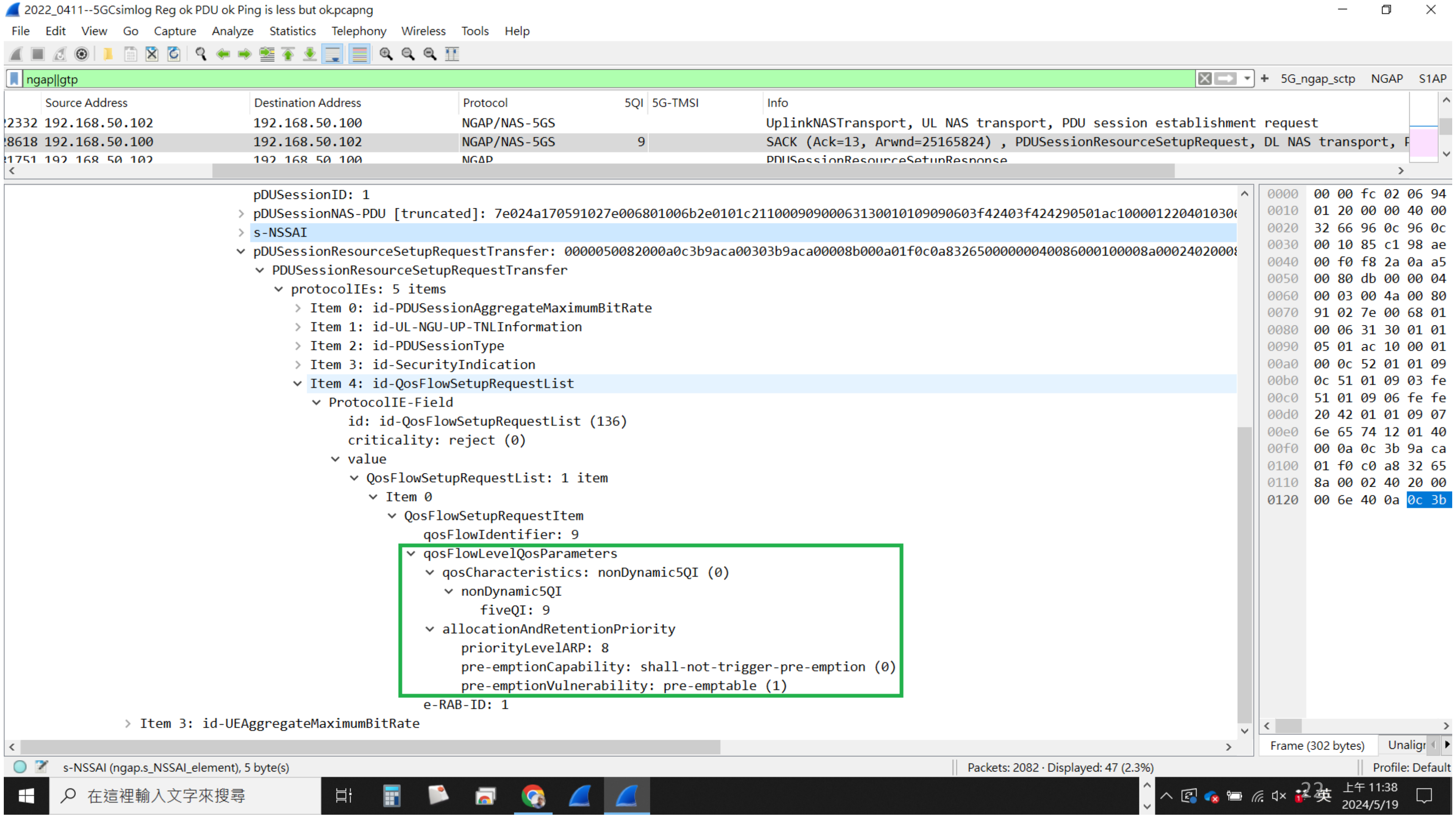
5G QoS Callflow – PDUsession Resource --2



5G QoS Callflow – PDUsession Resource --3



5G QoS Callflow – PDUsession Resource --4



5G QoS Callflow – translations for humen--2

AMF to gNB:

- UE AMBR
- PDUsession AMBR
- S-NSSAI of UE (which UE sent in Registration)
- UP gtp TEID, TLA
- QoS Flow Setup Request: QFI, e-RAB-ID, 5QI=9

5G QoS flows – Dynamic 5QI

5G NR Standardized QoS Identifier (5GQI) to QoS Characteristics Mapping

Packet Delay Budget (PDB)

BY AUTHOR · PUBLISHED JULY 25, 2019 · UPDATED JULY 25, 2019

Standardized 5G QoS Identifier (5QI) values are specified for services that are assumed to be frequently used in 5G networks and thus benefit from optimized signalling by using standardized QoS characteristics. Dynamically assigned 5QI values which require a signalling of QoS characteristics as part of the QoS profile can be used for services for which standardized 5QI values are not defined. The one-to-one mapping of standardized 5QI values to 5G QoS characteristics is specified in following table

5QI Value	Resource Type	Default Priority Level	Packet Delay Budget	Packet Error Rate	Default Maximum Data Burst Volume (NOTE 2)	Default Averaging Window	Example Services

https://www.techplayon.com/5g-nr-standardized-qos-identifier-5gqi-to-qos-characteristics-mapping/

5G QoS flows – Pre-emption

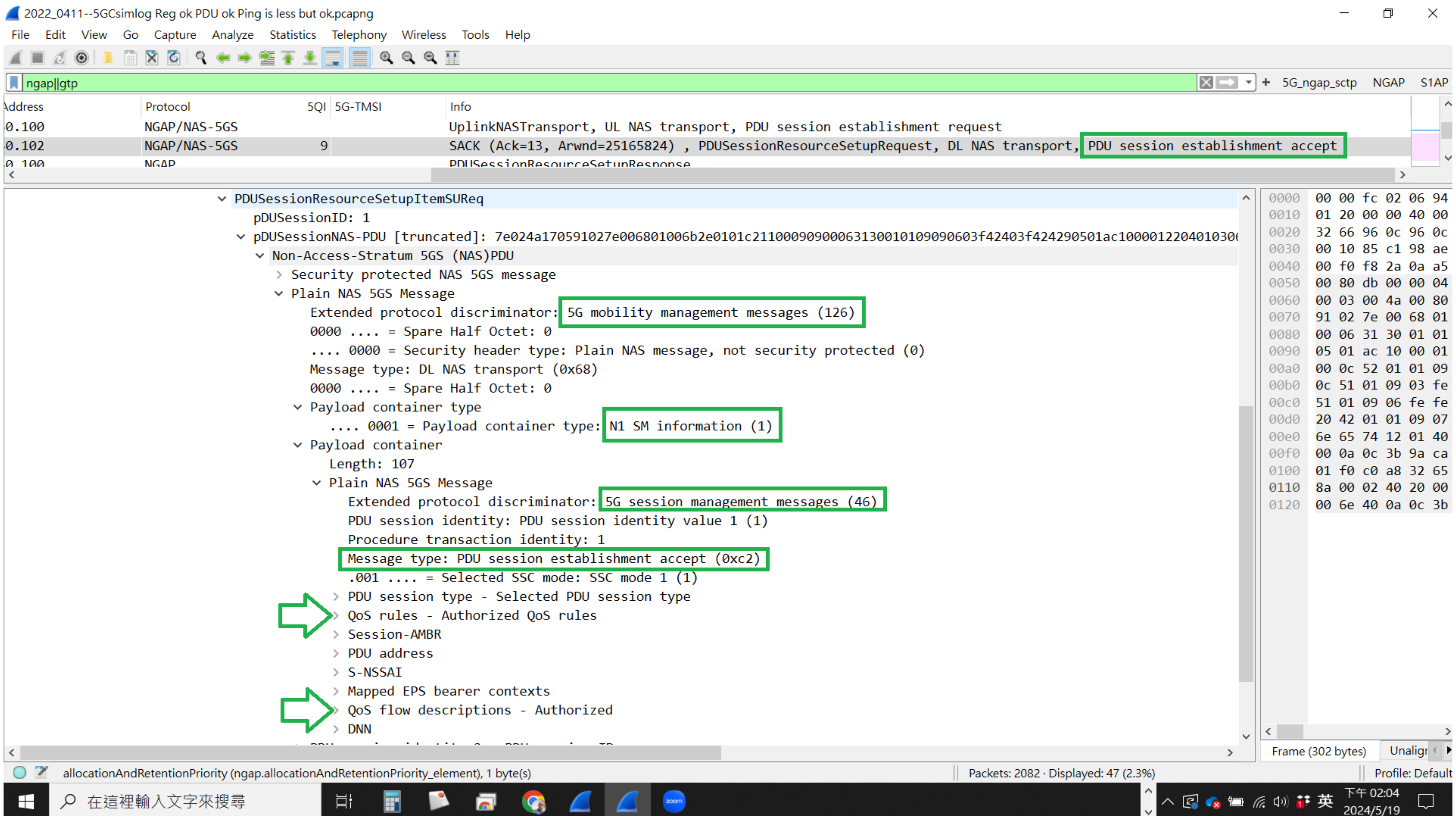
Allocation and Retention Priority (ARP)

The QoS parameter ARP contains information about the priority level, the pre-emption capability and the pre-emption vulnerability. The ARP priority level defines the relative importance of a resource request to allows in deciding whether a new QoS Flow may be accepted or needs to be rejected in the case of resource limitations (typically used for admission control of GBR traffic). It may also be used to decide which existing QoS Flow to pre-empt during resource limitations. ARP has following characteristics:

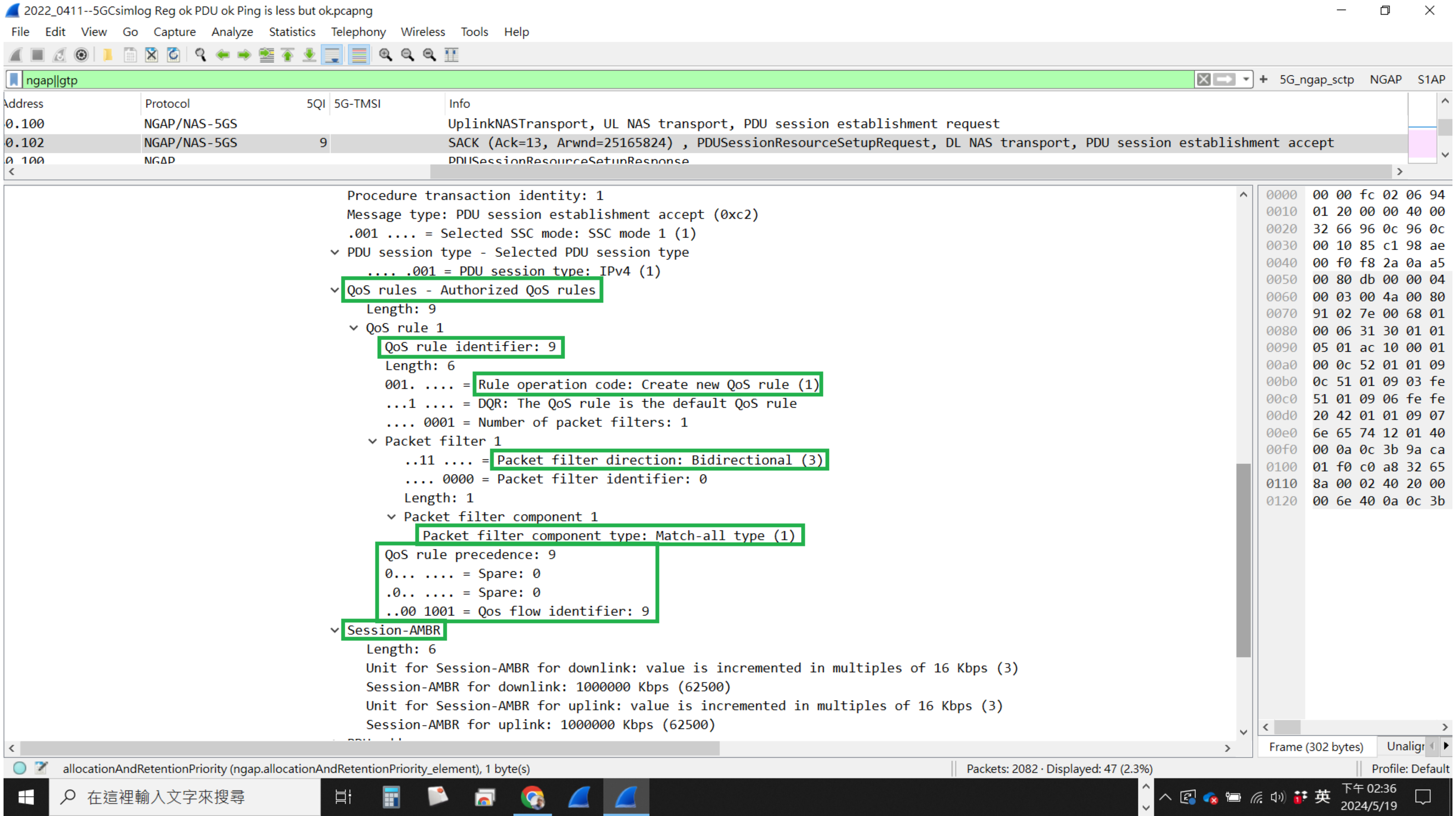
- The range of the ARP priority level is 1 to 15 with 1 as the highest level of priority
- The ARP priority levels 1-8 should only be assigned to resources for services that are authorized to receive prioritized treatment within an operator domain (i.e. that are authorized by the serving network)
- The ARP priority levels 9-15 may be assigned to resources that are authorized by the home network and thus applicable when a UE is roaming
- The ARP pre-emption capability defines whether a service data flow may get resources that were already assigned to another service data flow with a lower ARP priority level
- The ARP pre-emption capability and the ARP pre-emption vulnerability shall be either set to 'enabled' or 'disabled'
- The ARP pre-emption vulnerability defines whether a service data flow may lose the resources assigned to it in order to admit a service data flow with higher ARP priority level
- The ARP pre-emption vulnerability of the QoS Flow which the default QoS rule is associated with should be set appropriately to minimize the risk of unnecessary release of this QoS Flow

https://www.techplayon.com/5g-nr-qos-parameters/

5G QoS Callflow – PDUsessionEstAccept--1



5G QoS Callflow – PDUsessionEstAccept--2



5G QoS flows – Authorized QoS

3GPP TS 29.512 version 16.5.0 Release 16

4.2.6.6 Authorized QoS

4.2.6.6.1 General

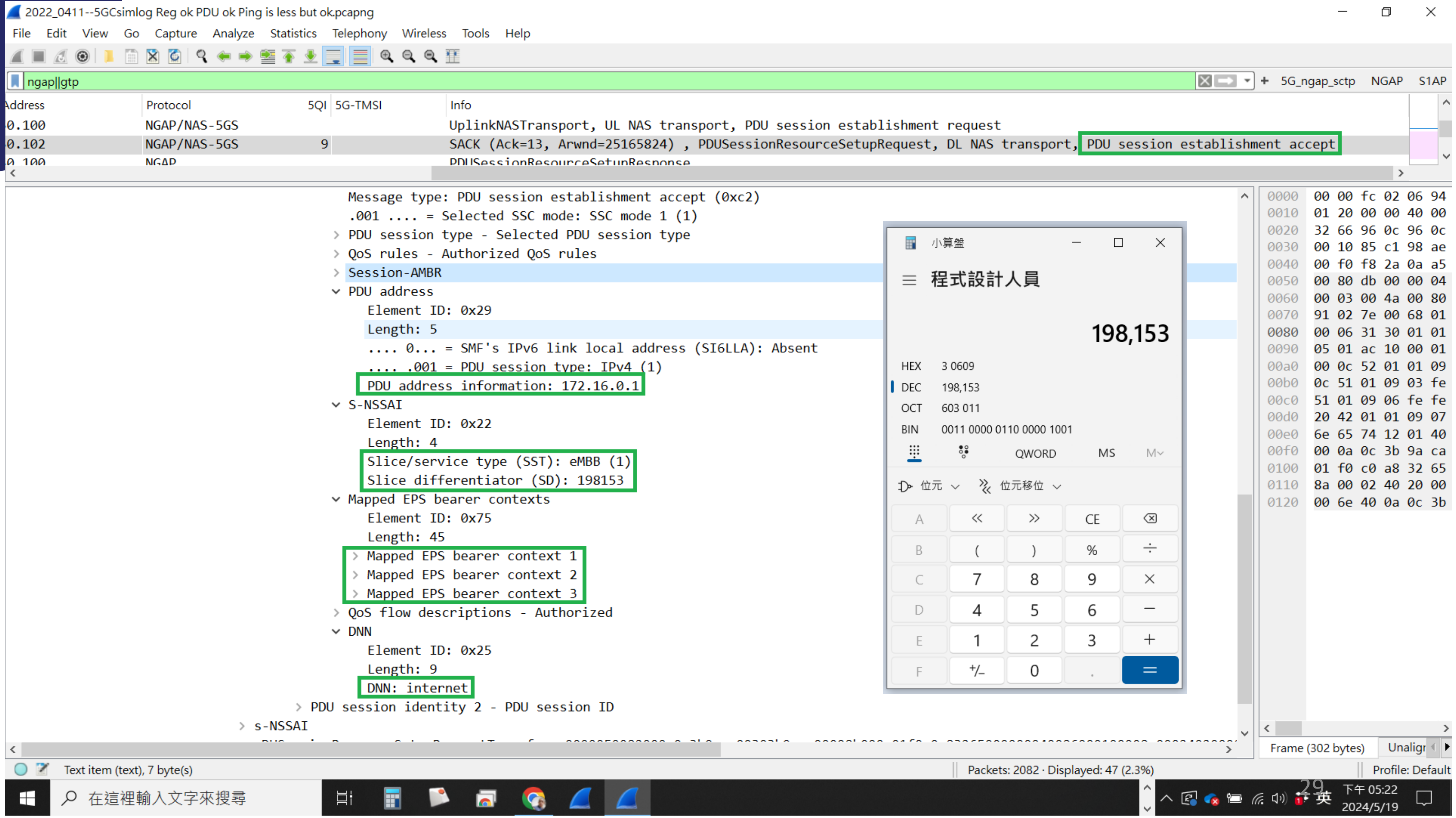
The PCF shall provision the authorized QoS. The authorized QoS may apply to a PCC rule or to a PDU session.

- When the authorized QoS applies to a PCC rule, it shall be provisioned within the corresponding PCC rule as defined in subclause 4.2.6.6.2.
- When the authorized QoS for a PCC rule with a GBR QCI is candidate for resource sharing an instruction on the allowed sharing may be provisioned as defined in subclause 4.2.6.2.8.
- When the authorized QoS applies to a PDU session, it shall be provisioned as defined in subclause 4.2.6.3.1.
- When the authorized QoS applies to the default QoS flow, it shall be provisioned as defined in subclause 4.2.6.3.1.
- When the authorized QoS applies to an explicitly signalled QoS Characteristics, it shall be provisioned as defined in subclause 4.2.6.6.3.
- When the authorized QoS applies to the Reflective QoS, it shall be provisioned as defined in subclause 4.2.6.5.7.

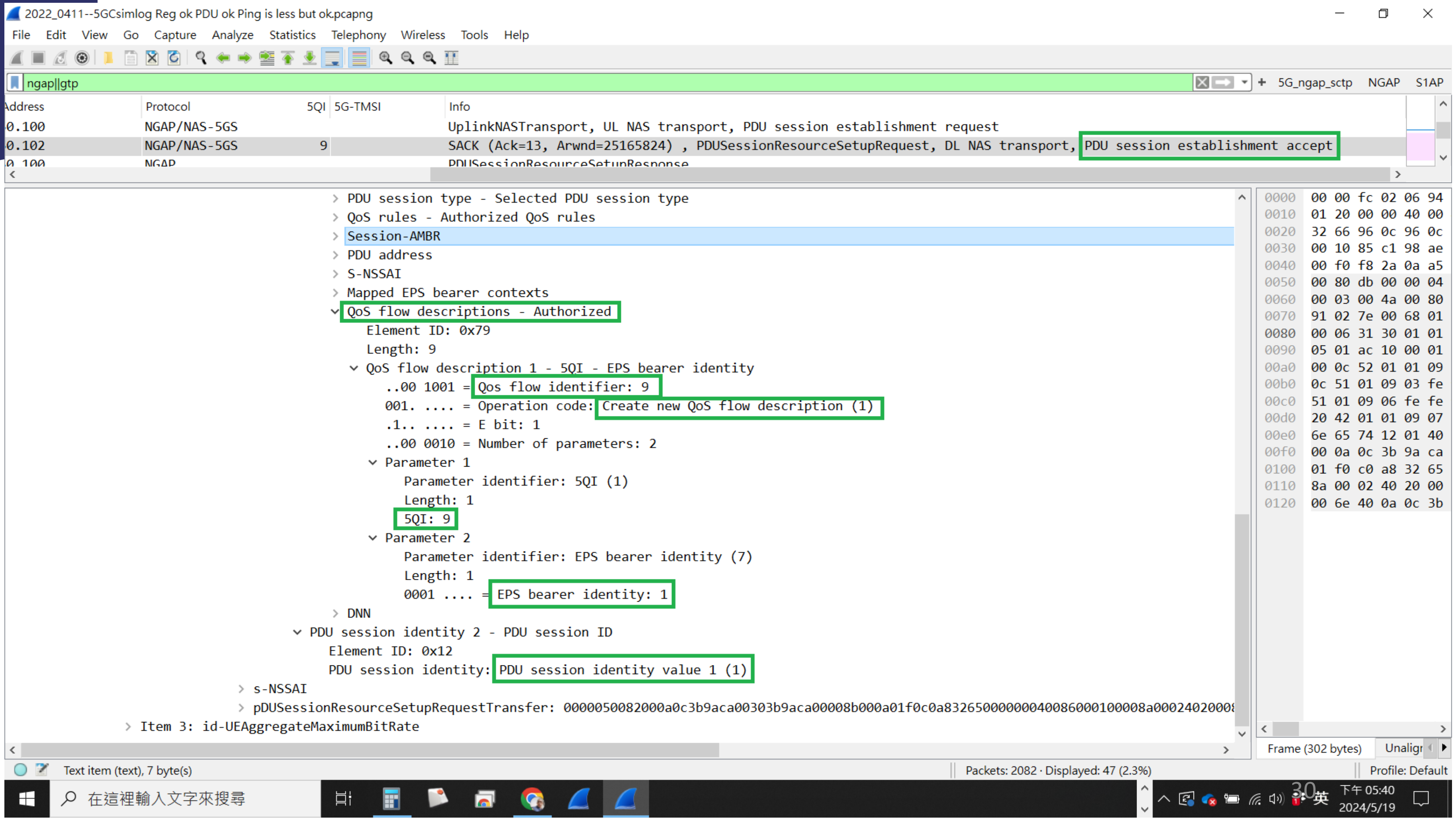
The authorized QoS provides appropriate values for the resources to be enforced. The authorized QoS for a PCC rule is a request for allocating the corresponding resources. The Provisioning of authorized QoS per PCC rule is a part of PCC rule provisioning procedure.

If the SMF cannot allocate any of the resources as authorized by the PCF, the SMF informs the PCF and acts as described in subclauses 4.2.3.16 and 4.2.4.15.

5G QoS Callflow – PDUsessionEstAccept--3



5G QoS Callflow – PDUsessionEstAccept--4

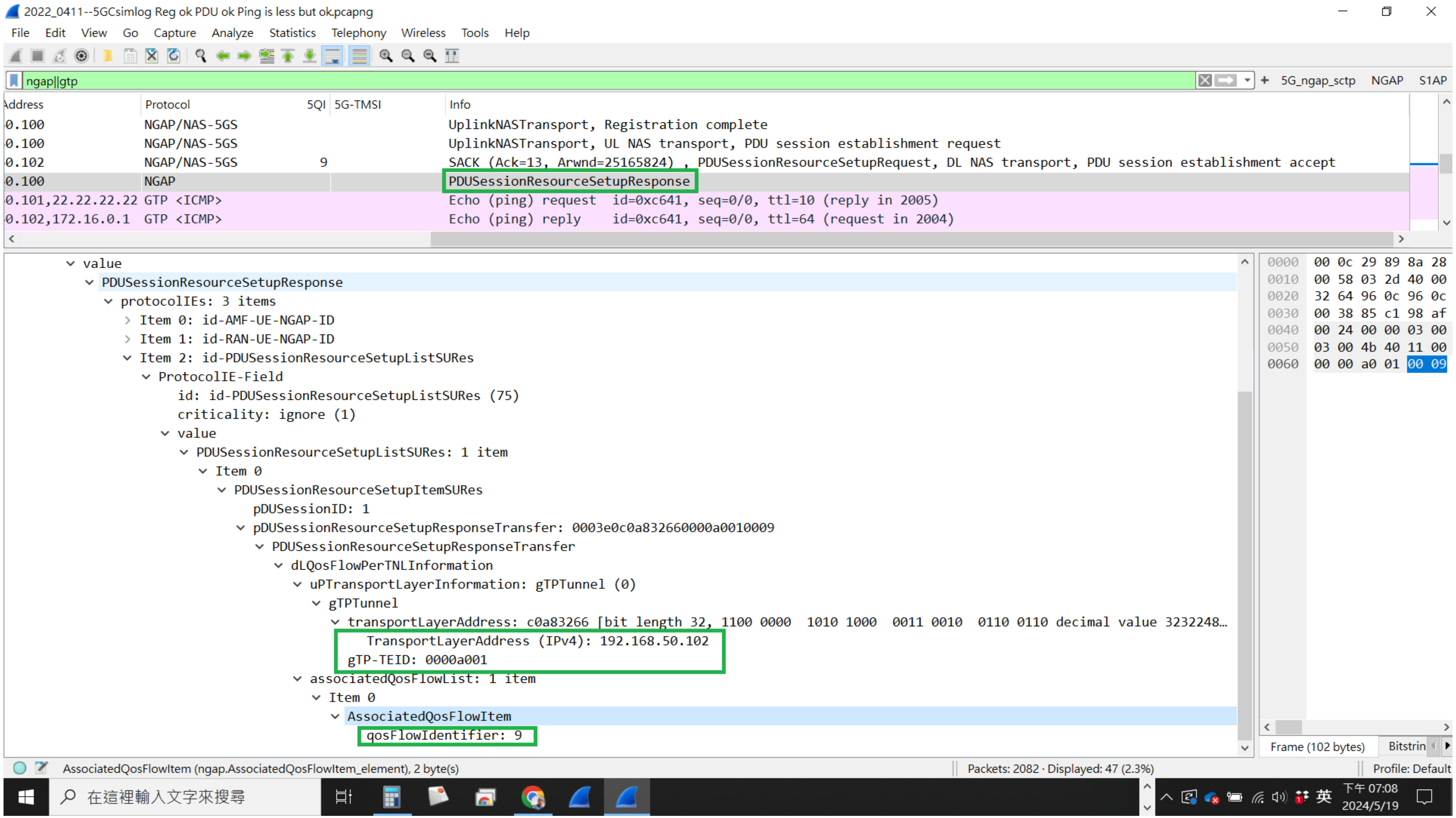


5G QoS Callflow – translations for humen--3

AMF to UE:

- Accept the PDUsessionEstRequest
- Send Authorized QoS rules to UE: to create new QoS rule, packet filter, QoS Flow ID
- Session AMBR
- PDU IP address
- S-NSSAI, DNN
- Send QoS Flow description to UE: QoS Flow ID, 5QI, EPS bearer ID
- PDUsession ID

5G QoS Callflow – PDUsession ResourceRSP



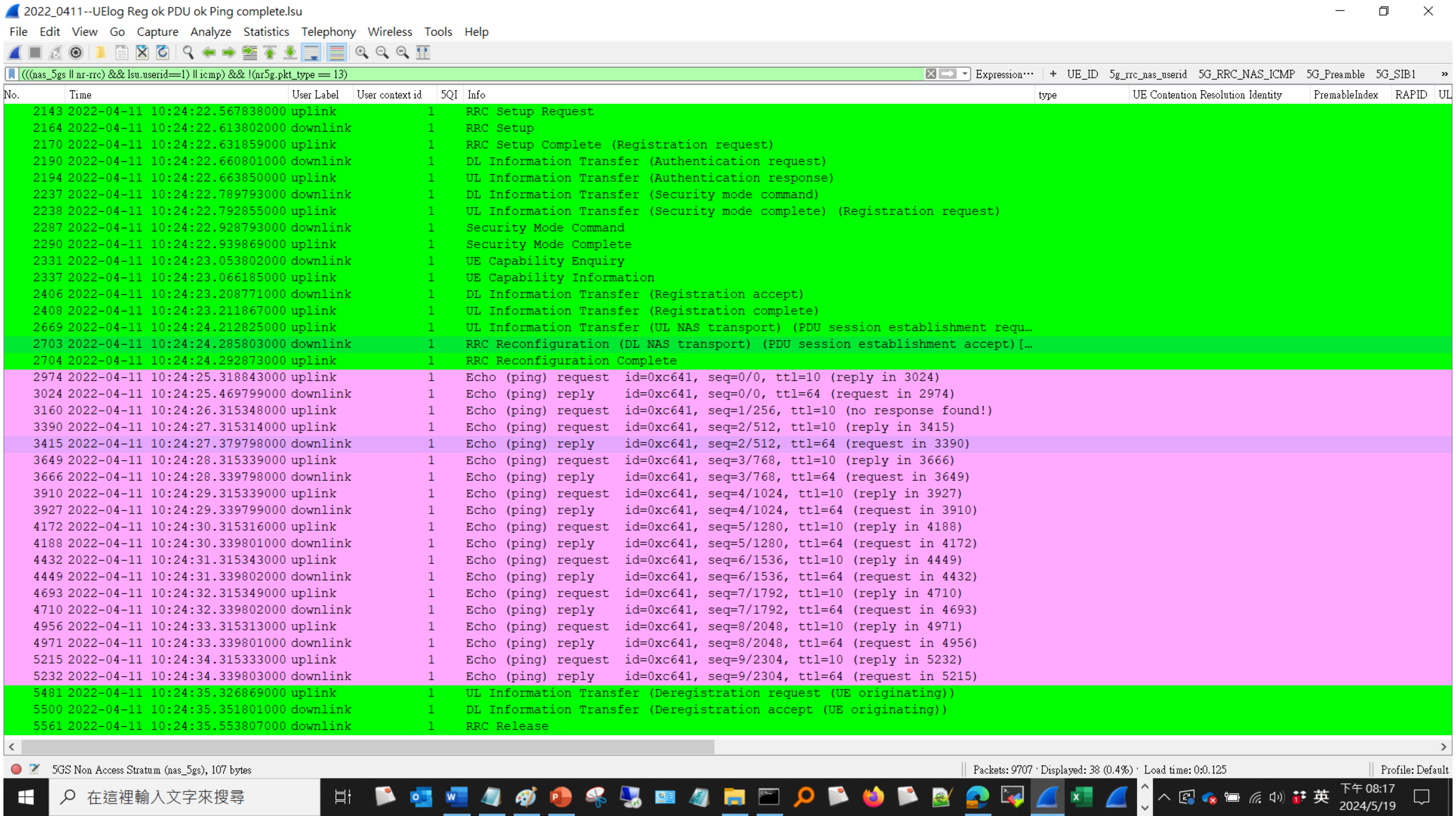
5G QoS Callflow – translations for humen--4

gNB to AMF:

- UP gtp TEID, TLA
- QoS Flow ID

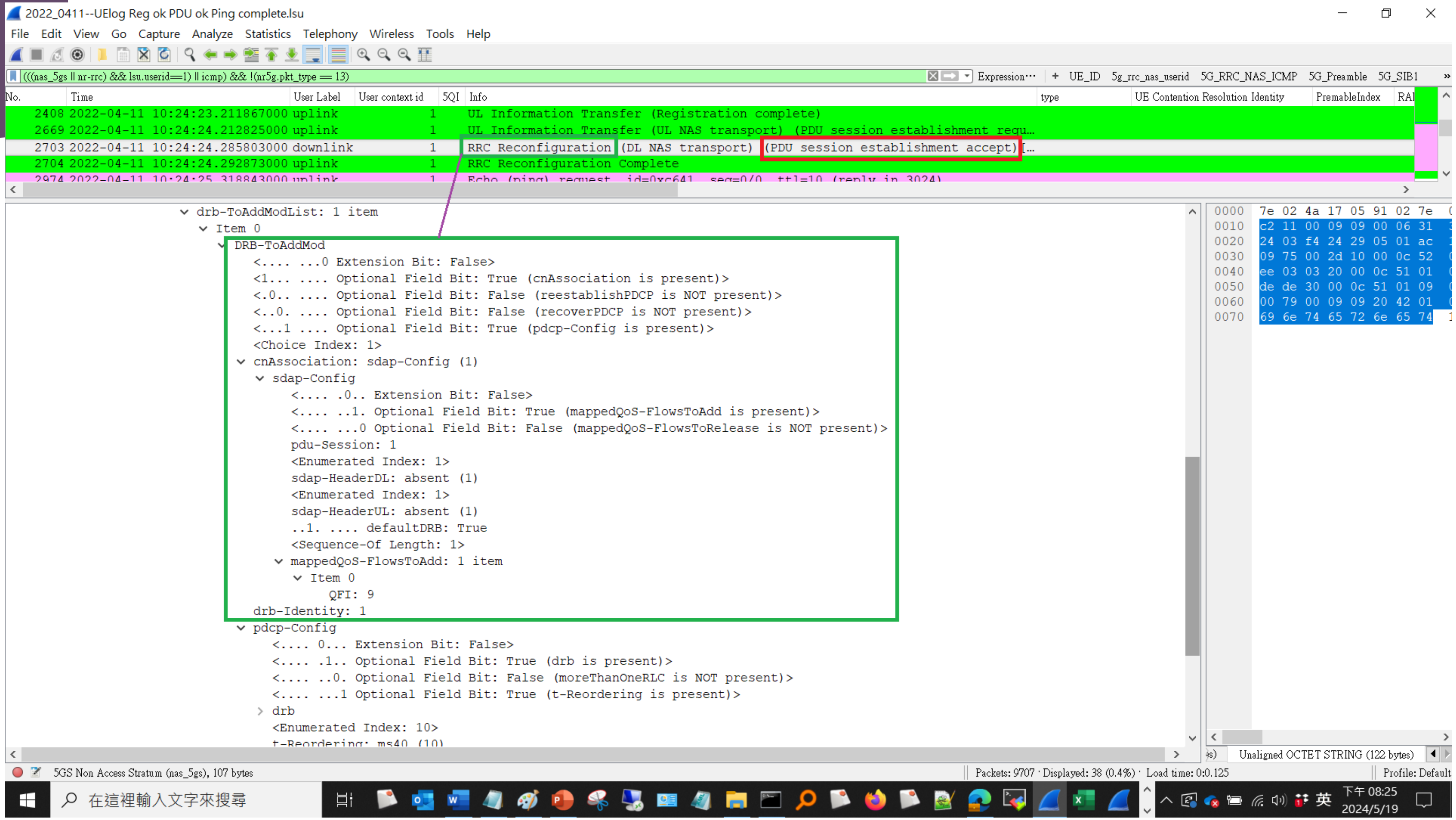
5G QoS Callflow – UElog

2022_0411--UElog Reg ok PDU ok Ping Complete.lsu



No.	Time	User Label	User context id	5QI	Info
2143	2022-04-11 10:24:22.567838000	uplink	1		RRC Setup Request
2164	2022-04-11 10:24:22.613802000	downlink	1		RRC Setup
2170	2022-04-11 10:24:22.631859000	uplink	1		RRC Setup Complete (Registration request)
2190	2022-04-11 10:24:22.660801000	downlink	1		DL Information Transfer (Authentication request)
2194	2022-04-11 10:24:22.663850000	uplink	1		UL Information Transfer (Authentication response)
2237	2022-04-11 10:24:22.789793000	downlink	1		DL Information Transfer (Security mode command)
2238	2022-04-11 10:24:22.792855000	uplink	1		UL Information Transfer (Security mode complete) (Registration request)
2287	2022-04-11 10:24:22.928793000	downlink	1		Security Mode Command
2290	2022-04-11 10:24:22.939869000	uplink	1		Security Mode Complete
2331	2022-04-11 10:24:23.053802000	downlink	1		UE Capability Enquiry
2337	2022-04-11 10:24:23.066185000	uplink	1		UE Capability Information
2406	2022-04-11 10:24:23.208771000	downlink	1		DL Information Transfer (Registration accept)
2408	2022-04-11 10:24:23.211867000	uplink	1		UL Information Transfer (Registration complete)
2669	2022-04-11 10:24:24.212825000	uplink	1		UL Information Transfer (UL NAS transport) (PDU session establishment requ…
2703	2022-04-11 10:24:24.285803000	downlink	1		RRC Reconfiguration (DL NAS transport) (PDU session establishment accept)[…
2704	2022-04-11 10:24:24.292873000	uplink	1		RRC Reconfiguration Complete
2974	2022-04-11 10:24:25.318843000	uplink	1		Echo (ping) request id=0xc641, seq=0/0, ttl=10 (reply in 3024)
3024	2022-04-11 10:24:25.469799000	downlink	1		Echo (ping) reply id=0xc641, seq=0/0, ttl=64 (request in 2974)
3160	2022-04-11 10:24:26.315348000	uplink	1		Echo (ping) request id=0xc641, seq=1/256, ttl=10 (no response found!)
3390	2022-04-11 10:24:27.315314000	uplink	1		Echo (ping) request id=0xc641, seq=2/512, ttl=10 (reply in 3415)
3415	2022-04-11 10:24:27.379798000	downlink	1		Echo (ping) reply id=0xc641, seq=2/512, ttl=64 (request in 3390)
3649	2022-04-11 10:24:28.315339000	uplink	1		Echo (ping) request id=0xc641, seq=3/768, ttl=10 (reply in 3666)
3666	2022-04-11 10:24:28.339798000	downlink	1		Echo (ping) reply id=0xc641, seq=3/768, ttl=64 (request in 3649)
3910	2022-04-11 10:24:29.315339000	uplink	1		Echo (ping) request id=0xc641, seq=4/1024, ttl=10 (reply in 3927)
3927	2022-04-11 10:24:29.339799000	downlink	1		Echo (ping) reply id=0xc641, seq=4/1024, ttl=64 (request in 3910)
4172	2022-04-11 10:24:30.315316000	uplink	1		Echo (ping) request id=0xc641, seq=5/1280, ttl=10 (reply in 4188)
4188	2022-04-11 10:24:30.339801000	downlink	1		Echo (ping) reply id=0xc641, seq=5/1280, ttl=64 (request in 4172)
4432	2022-04-11 10:24:31.315343000	uplink	1		Echo (ping) request id=0xc641, seq=6/1536, ttl=10 (reply in 4449)
4449	2022-04-11 10:24:31.339802000	downlink	1		Echo (ping) reply id=0xc641, seq=6/1536, ttl=64 (request in 4432)
4693	2022-04-11 10:24:32.315349000	uplink	1		Echo (ping) request id=0xc641, seq=7/1792, ttl=10 (reply in 4710)
4710	2022-04-11 10:24:32.339802000	downlink	1		Echo (ping) reply id=0xc641, seq=7/1792, ttl=64 (request in 4693)
4956	2022-04-11 10:24:33.315313000	uplink	1		Echo (ping) request id=0xc641, seq=8/2048, ttl=10 (reply in 4971)
4971	2022-04-11 10:24:33.339801000	downlink	1		Echo (ping) reply id=0xc641, seq=8/2048, ttl=64 (request in 4956)
5215	2022-04-11 10:24:34.315333000	uplink	1		Echo (ping) request id=0xc641, seq=9/2304, ttl=10 (reply in 5232)
5232	2022-04-11 10:24:34.339803000	downlink	1		Echo (ping) reply id=0xc641, seq=9/2304, ttl=64 (request in 5215)
5481	2022-04-11 10:24:35.326869000	uplink	1		UL Information Transfer (Deregistration request (UE originating))
5500	2022-04-11 10:24:35.351801000	downlink	1		DL Information Transfer (Deregistration accept (UE originating))
5561	2022-04-11 10:24:35.553807000	downlink	1		RRC Release

5G QoS Callflow – RRC Reconfiguration



5G QoS Callflow – translations for humen--5

gNB to UE:

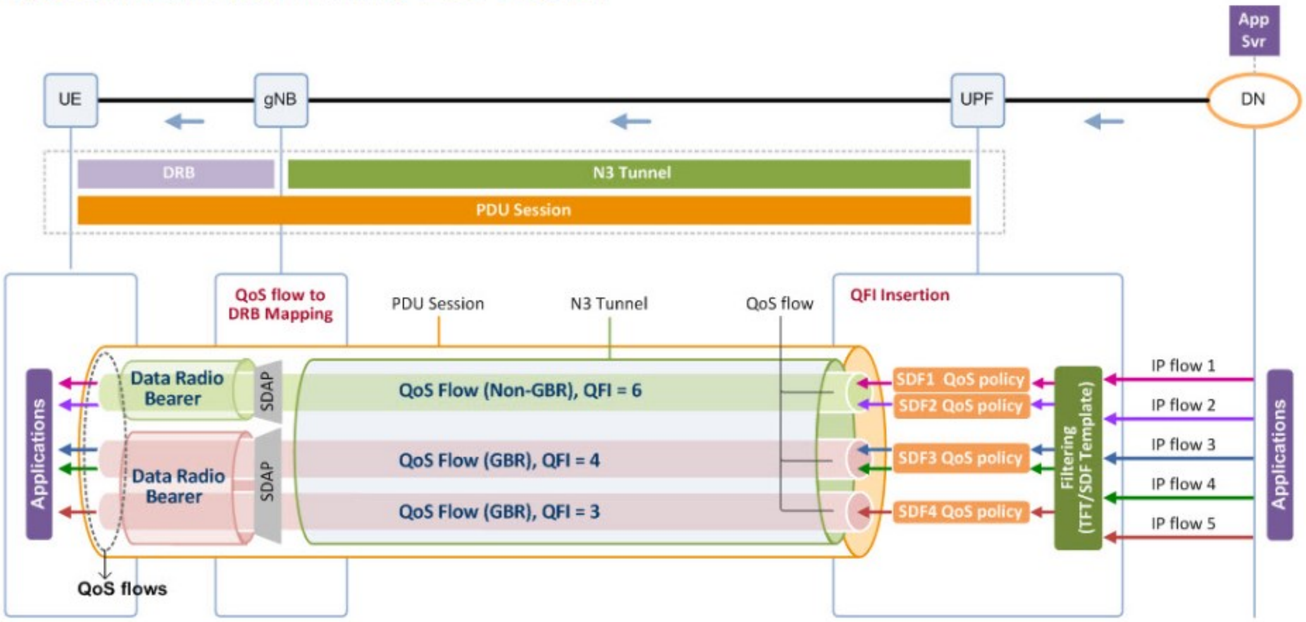
- mappedQoS-FlowtoAdd
- QoS Flow ID

5G QoS flows -- DL

https://netmanias.com/en/post/oneshot/14105/5g/5g-qos

NETMANIAS ONE-SHOT | 5G QoS - Downlink | February 7, 2019

The QoS differentiation within a PDU session



5QI	: 5G QoS Identifier
ARP	: Allocation and Retention Priority
GFBR	: Guaranteed Flow Bit Rate
MFBR	: Maximum Flow Bit Rate
PDB	: Packet Delay Budget
PER	: Packet Error Rate
QFI	: QoS Flow Identifier
RQA	: Reflective QoS Attribute

QoS Flow type		QoS Flow parameters
GBR flow	Non-GBR flow	5QI
		ARP
		RQA
		GFBR
		MFBR
		Notification Control
		Maximum Packet Loss Rate

5QI

- Resource Type*
- Default Priority Level
- PDB
- PER
- Default Maximum Data Burst Volume
- Default Averaging Window

* GBR, non-GBR or delay critical GBR

5QI Characteristics

- Example of GBR

5G QoS Identifier | 5QI Table

Following table mentions 5QI values and their corresponding QoS characteristics mapping.

5QI Value	Resource Type	Default Priority Level	Packet Delay Budget	Packet Error Rate	Default Maximum Data Burst Volume	Default Averaging Window	Example Services
1	GBR (guaranteed flow bit rate)	20	100 ms	10^{-2}	N/A	2000 ms	Conversational Voice
2	GBR	40	150 ms	10^{-3}	N/A	2000 ms	Conversational Video (Live Streaming)
3	GBR	30	50 ms	10^{-3}	N/A	2000 ms	Real time gaming, V2X messages
4	GBR	50	300 ms	10^{-6}	N/A	2000 ms	Non-conversational Video (Buffered Streaming)
65	GBR	7	75 ms	10^{-2}	N/A	2000 ms	Mission Critical user plane Push to Talk voice (e.g. MCPTT)
66	GBR	20	100 ms	10^{-2}	N/A	2000 ms	Non-Mission Critical user plane Push to Talk Voice
67	GBR	15	100 ms	10^{-3}	N/A	2000 ms	Mission Critical Video user plane

https://www.rfwireless-world.com/5G/5G-QoS-Identifier-5QI.html

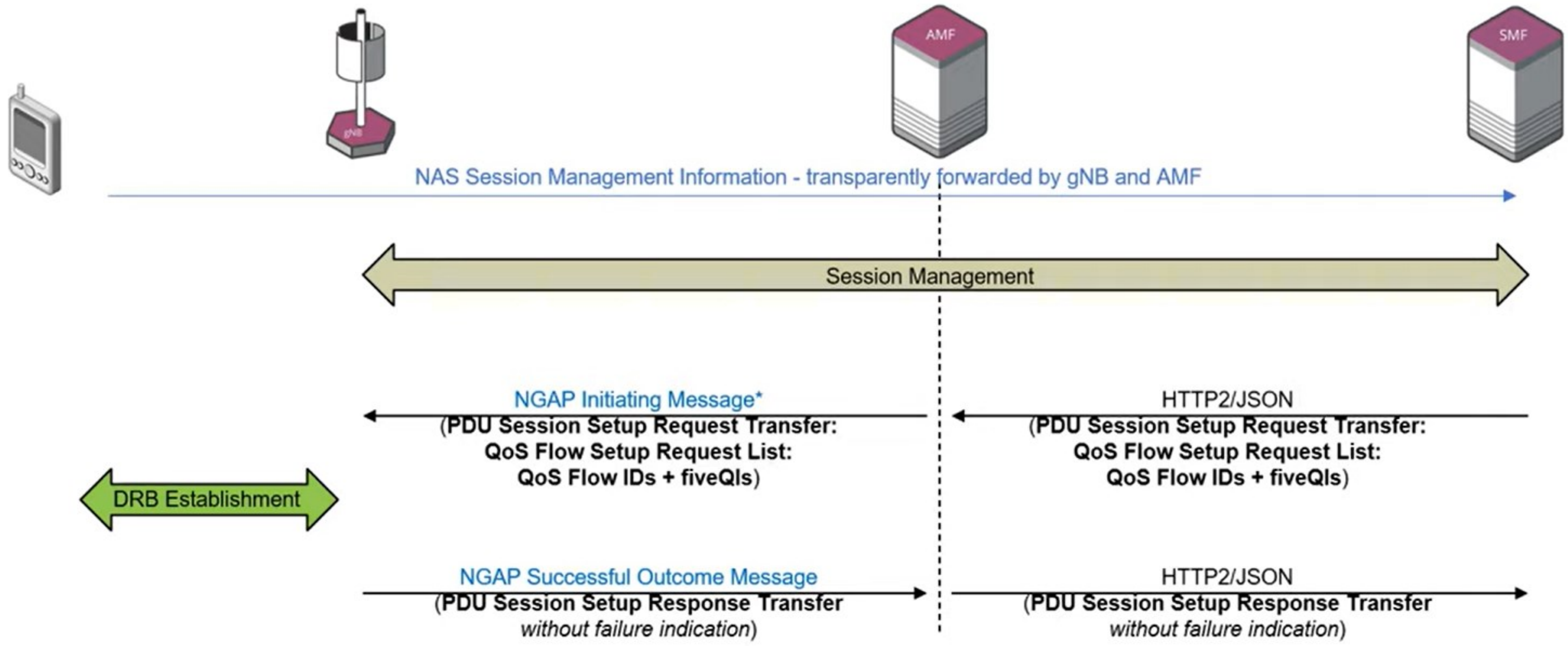
5QI Characteristics

- Example of NGBR

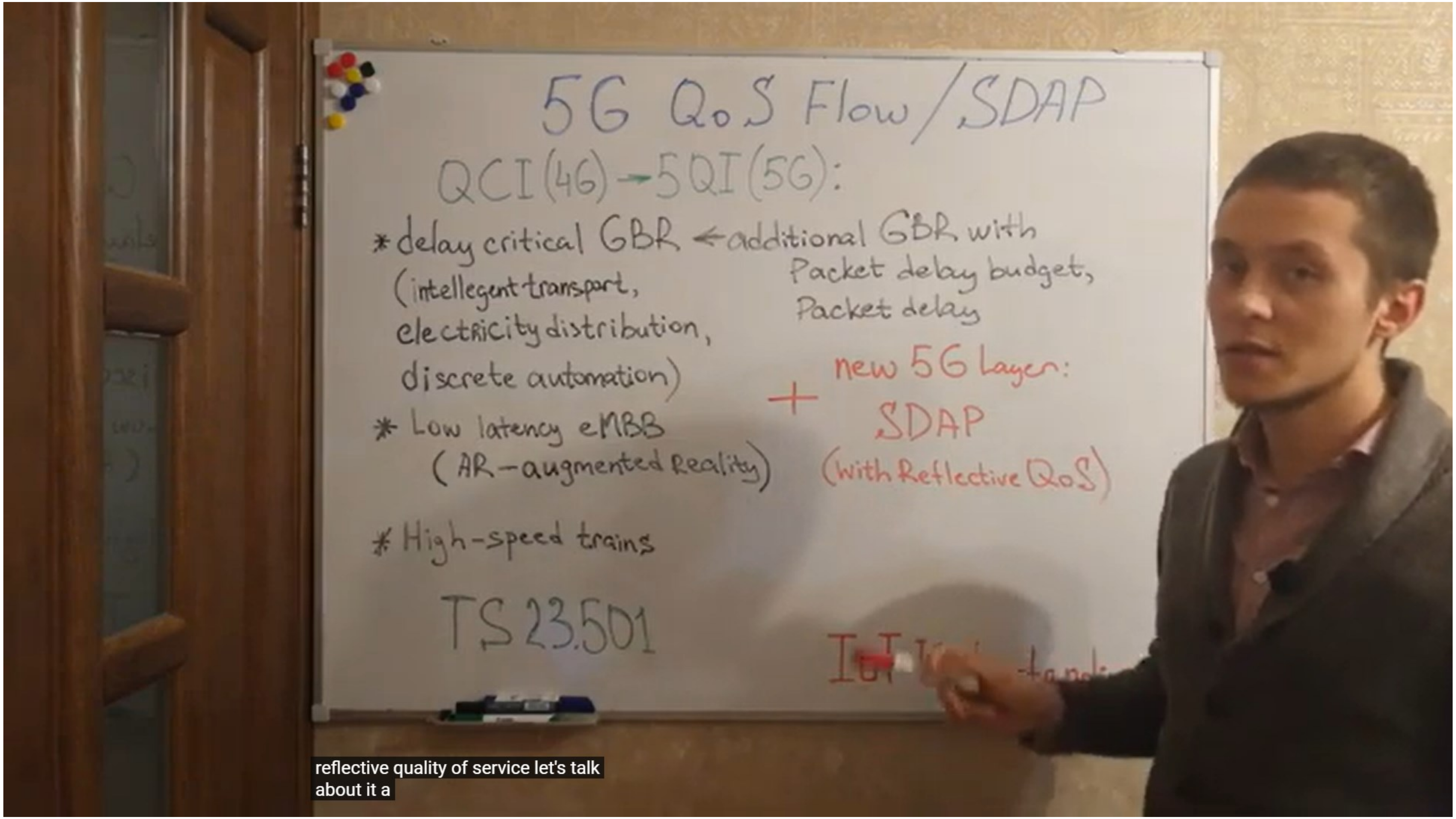
74	GBR	56	500 ms	$10^{[illegible]}$	N/A	2000 ms	"Live" Uplink Streaming
76	GBR	56	500 ms	10^{-4}	N/A	2000 ms	"Live" Uplink Streaming
5	Non-GBR	10	100 ms	10^{-6}	N/A	N/A	IMS Signalling
6	Non-GBR	60	300 ms	10^{-6}	N/A	N/A	Video (Buffered Streaming) TCP-based (e.g. www, e-mail, chat, FTP, p2p file sharing, progressive video etc.)
7	Non-GBR	70	100 ms	10^{-3}	N/A	N/A	Voice, Video (Live Streaming), Interactive Gaming
8	Non-GBR	80	300 ms	10^{-6}	N/A	N/A	Video (Buffered Streaming) TCP-based (e.g. www, e-mail, chat, FTP, p2p file sharing, progressive video etc.)
9	Non-GBR	90	-As above-	-As above-	-As above-	-As above-	-As above-
69	Non-GBR	5	60 ms	10^{-6}	N/A	N/A	Mission Critical Delay Sensitive Signalling (e.g. MC-PTT Signalling)
70	Non-GBR	55	200 ms	10^{-6}	N/A	N/A	Mission Critical Data (e.g. example services are the same as 5QI 6/8/9)

https://www.rfwireless-world.com/5G/5G-QoS-Identifier-5QI.html https://youtu.be/fnii_fA1W4s?si=lJIrD1JGTXMmE8mk

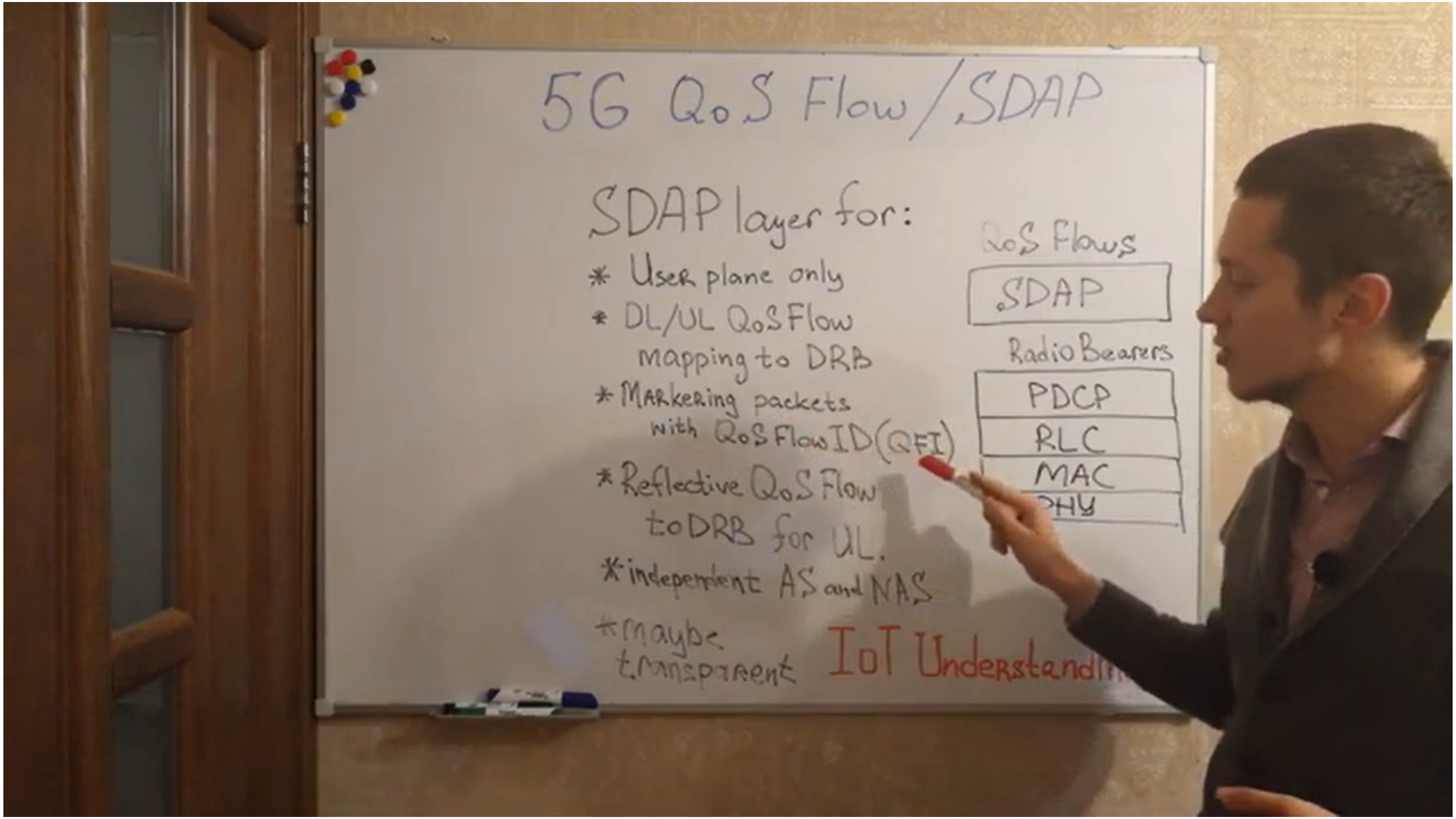
QoS Flow Setup: gNB - AMF - SMF



* This can be:
NGAP Initial Context Setup Request
NGAP PDU Session Resource Setup Request
NGAP PDU Session Resource Modify Request
NGAP Handover Request (with different success/failure Response message)

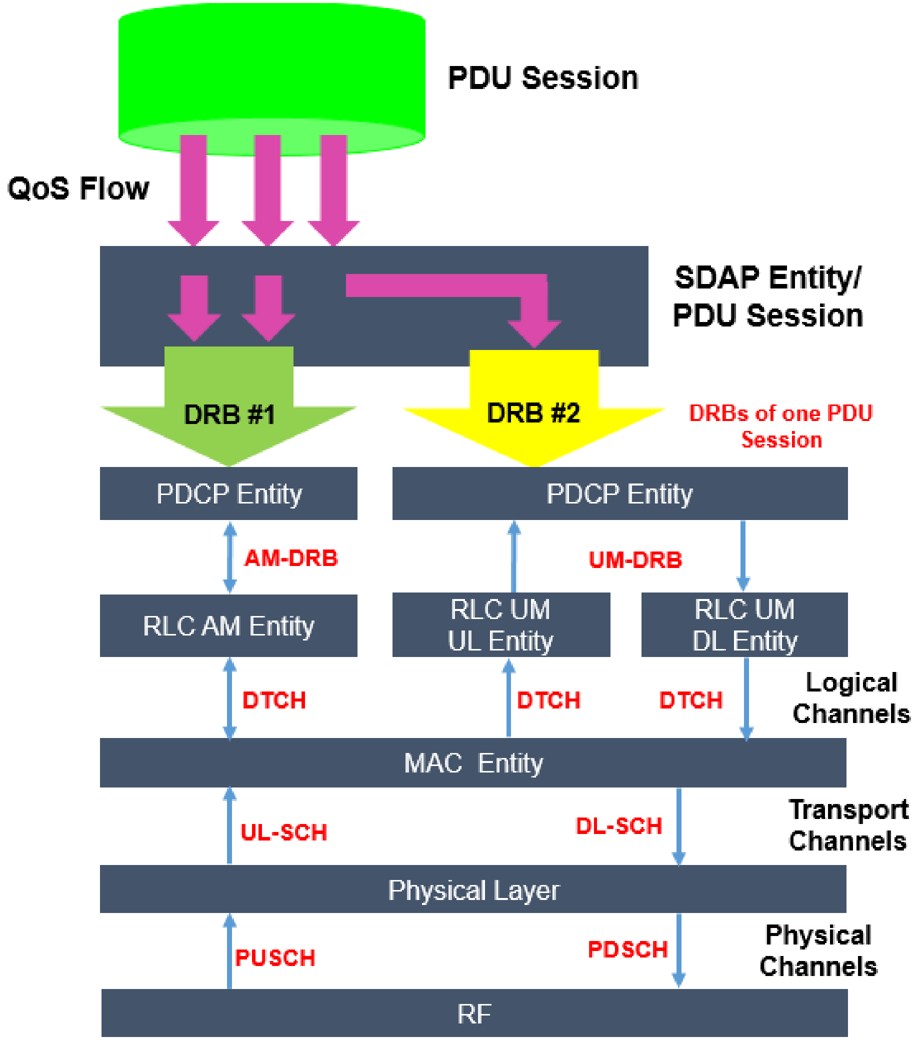


https://youtu.be/XTEQDhmy8SY?si=H41JBR5mSew2R0yE



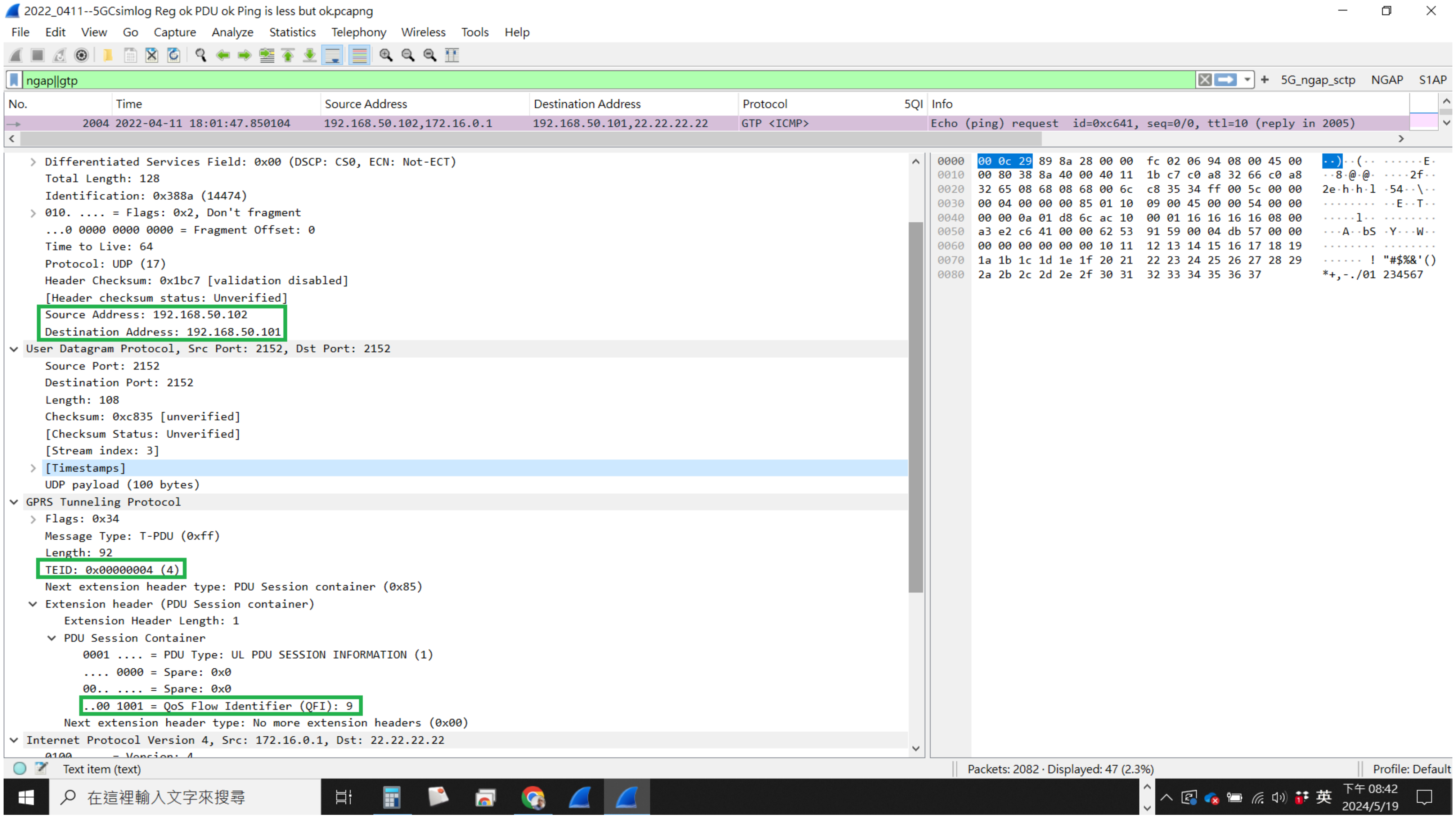
https://youtu.be/XTEQDhmy8SY?si=H41JBR5mSew2R0yE

Important pages



https://www.techplayon.com/5g-nr-sdap-service-data-adaption-protocol/

5G QoS Callflow – Ping

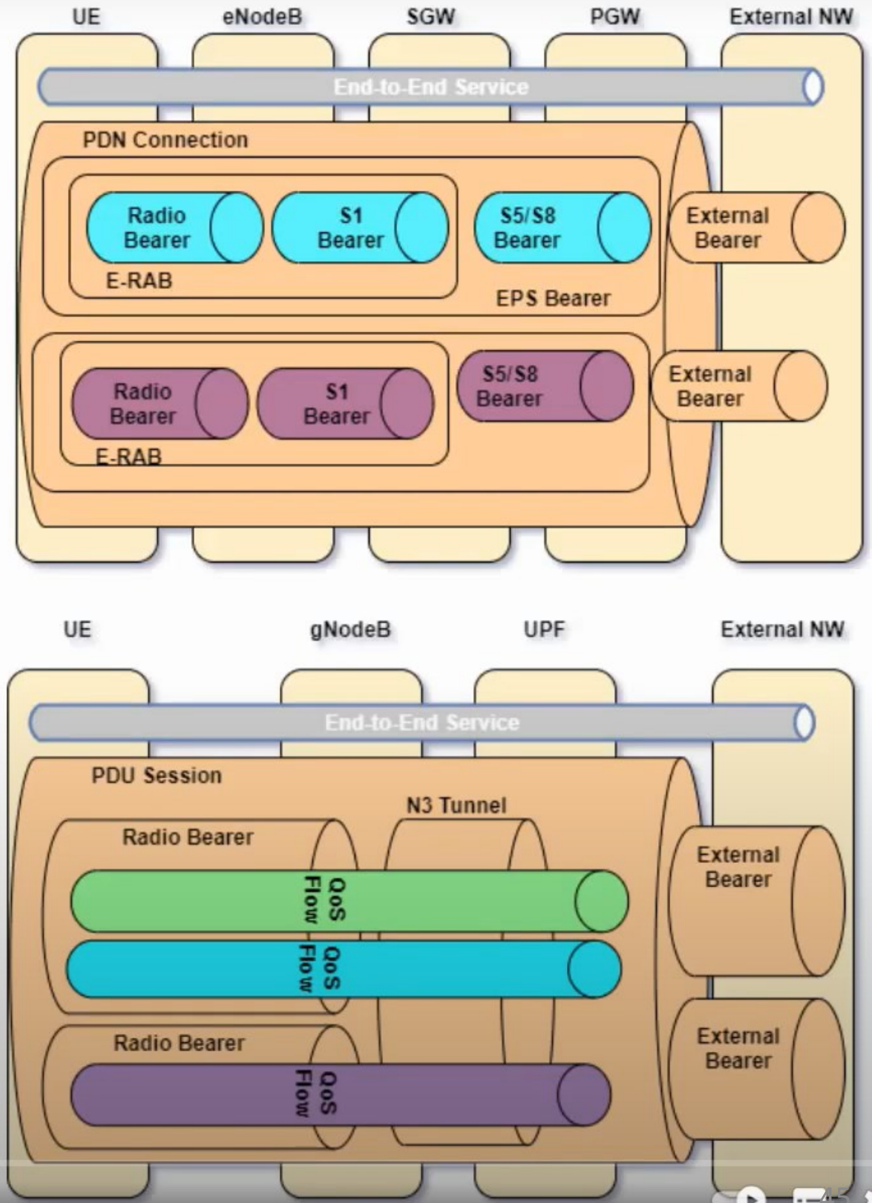


QoS (Quality of Service) in 5G

https://youtu.be/GUNQjwjEwQM?si=J5WmMfoxtgIb8-Lf

5G QoS

- In 5G QoS framework is based on QoS Flows which is the finest granularity of QoS differentiation.
- Each QoS Flow is identified by a QoS Flow ID.
- NG-RAN can map multiple QoS Flow IDs to one Radio Bearer or can have a 1:1 relationship. It depends on NG-RAN implementation.
- In 4G we had a 1:1 relationship between QCI and EPS bearers.
- 3GPP has standardized QoS Flow IDs like QCI for various services. More on this later.
- Also notice there is one N3 tunnel between eNB and UPF unlike multiple S5/S8 bearers in LTE.



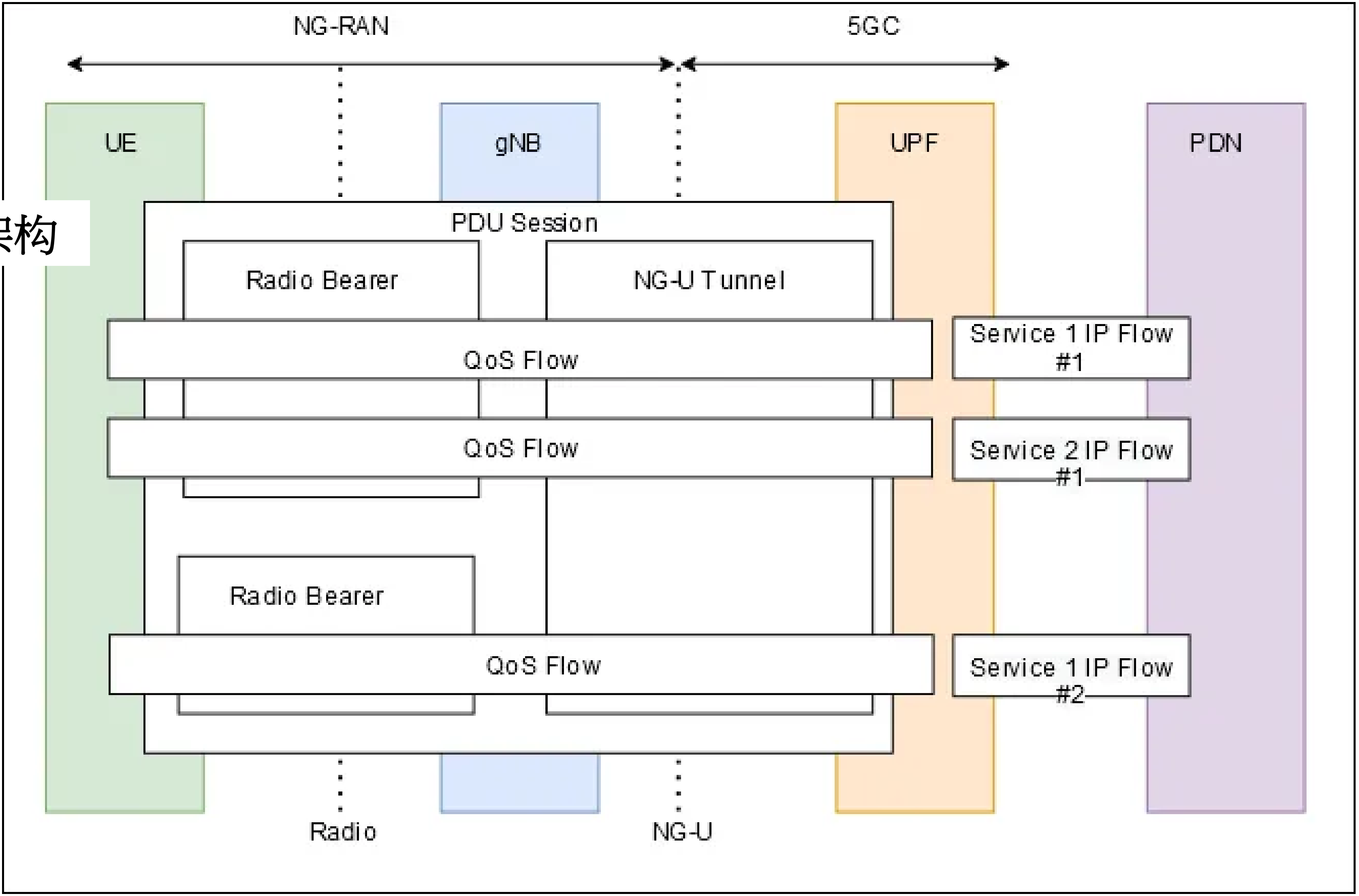
0:11 / 16:57

5G QoS vs 4G QoS

https://mp.weixin.qq.com/s/3fppz-mq_8LZZ_7IJllPhw

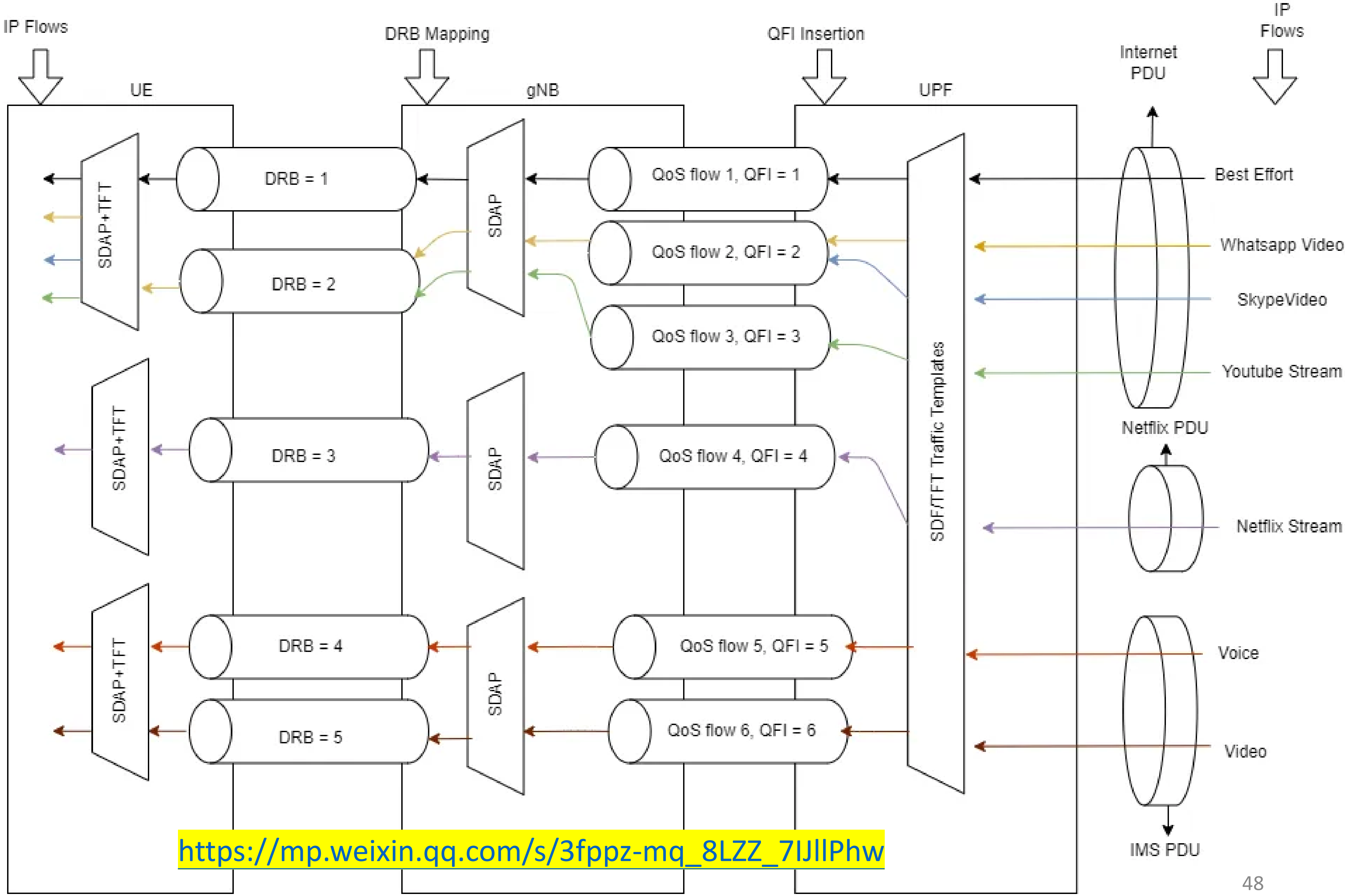
Parameter	5G	4G
QoS Identifier	5G QI (QoS Identifier)	QCI (QoS Class Identifier)
IP Data Flow	QoS Flow	EPC Bearer
Flow/Bearer ID	QFI (QoS Flow Identifier	EBI
Reflective QoS	RQI (Reflective QoS ID)	NA
Data Session	PDU Session	PDN Connection

5G QoS架构



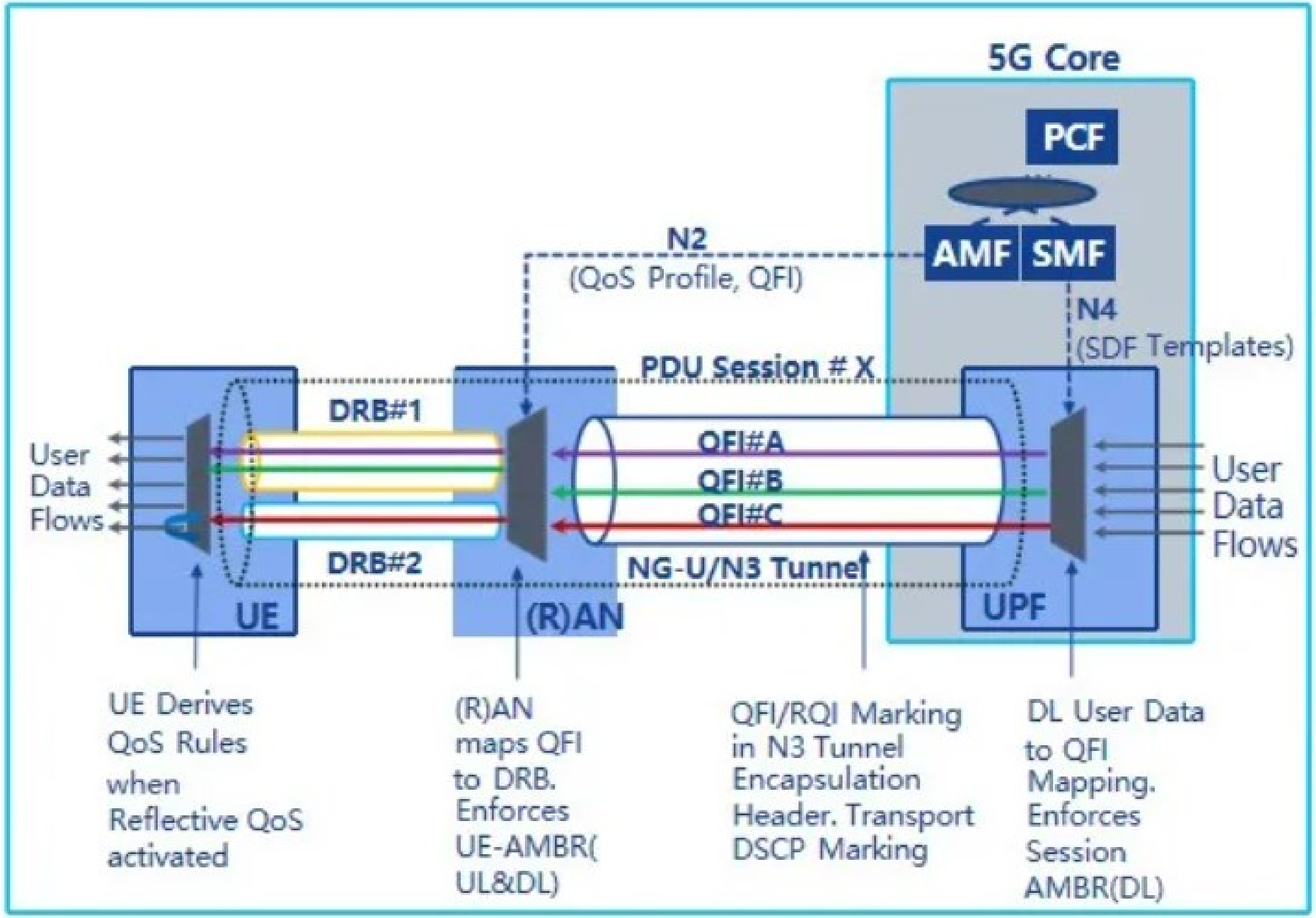
https://mp.weixin.qq.com/s/3fppz-mq_8LZZ_7IJllPhw

5G QoS示例



https://mp.weixin.qq.com/s/3fppz-mq_8LZZ_7IJlIPhw https://mp.weixin.qq.com/s/SLsT878qan6qXwQiwELR6g

下行QoS流处理



UPF将服务数据流（SDF）映射到QoS流

UPF执行Session AMBR并且还可以对PDU计数以支持计费

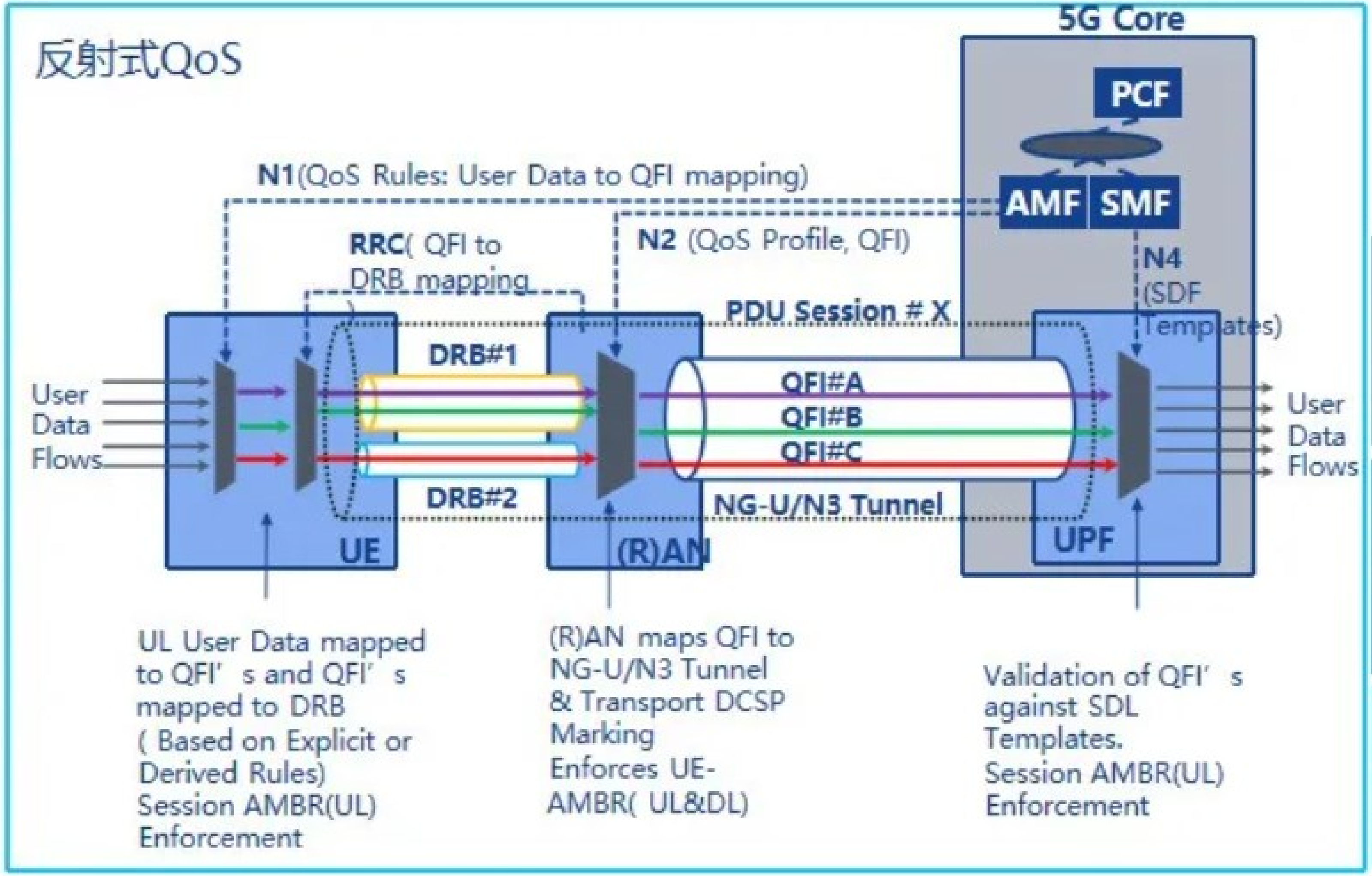
UPF在5GC和(R)AN之间的单个隧道中传输PDU会话的数据包，UPF封装报头中的用户平面标记（包括用于非3GPP接入的5QI）。另外，UPF可以在封装报头中包括对反射式QoS激活的指示。

(R)AN根据QFI和相关联的5G QoS特性和参数将来自QoS流的PDU映射到特定接入的资源。

如果应用反射QoS，则UE创建新的派生QoS规则。派生QoS规则中的分组过滤器来自下行分组包（即，报头），并且派生的QoS规则的用户面标记来自于下行分组包的用户面标记。

https://mp.weixin.qq.com/s/SLsT878qan6qXwQiwELR6g

上行QoS流处理



UE使用存储的QoS规则来确定SDF与QoS流之间的映射。 UE基于RAN提供的映射，使用与QoS流相对应的接入特定资源来传输上行PDU。

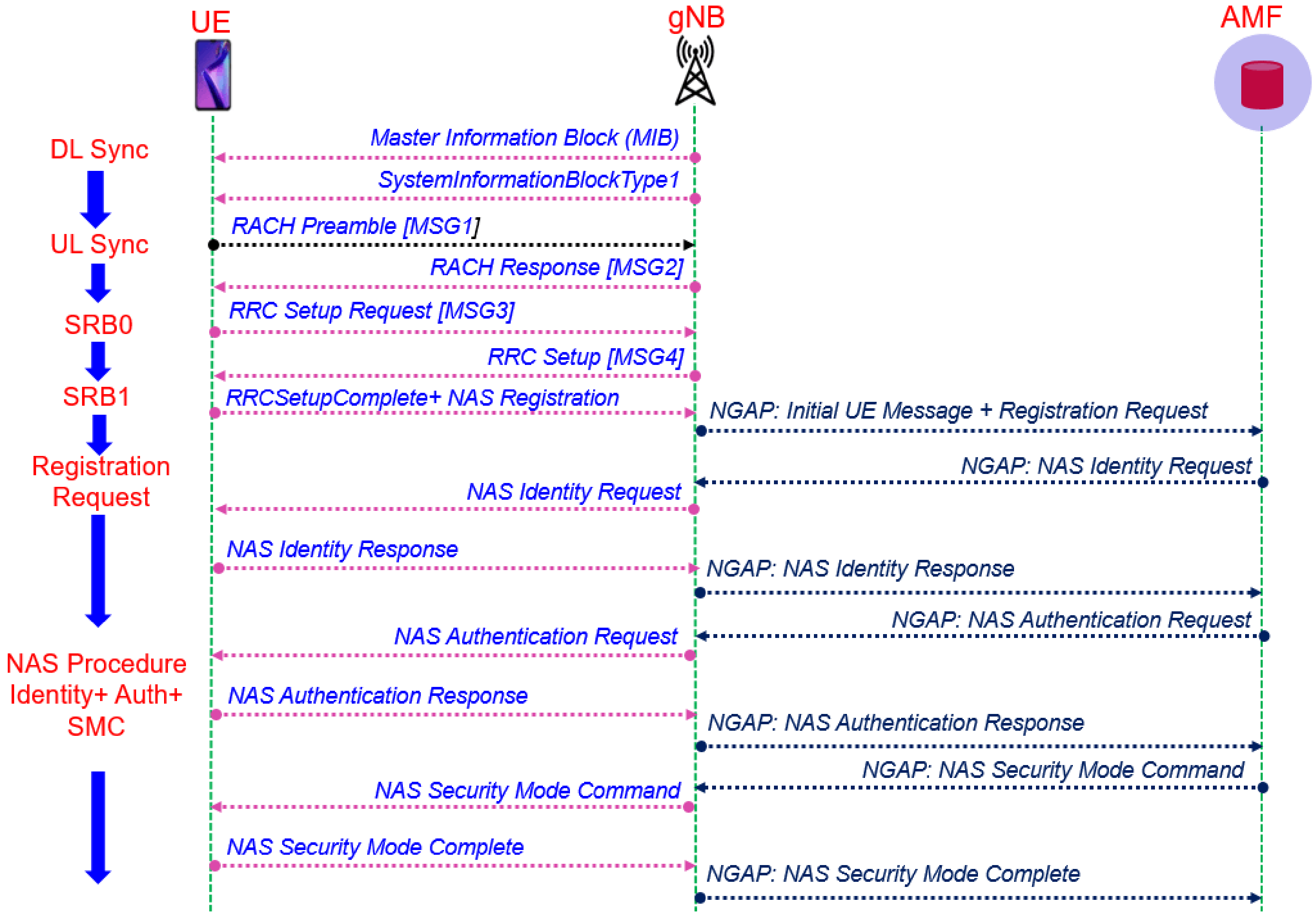
(R)AN经由N3隧道向UPF传输PDU。在从(R)AN向CN传递上行分组包时，(R)AN可确定QFI值（该值包含在UL PDU的封装报头中），并选择N3隧道。

UPF验证UL PDU中的QFI是否与提供给UE或由UE隐式派生（例如，在采用反射式QoS的情况下）的QoS规则相一致。

UPF执行session AMBR并对PDU计数以进行计费

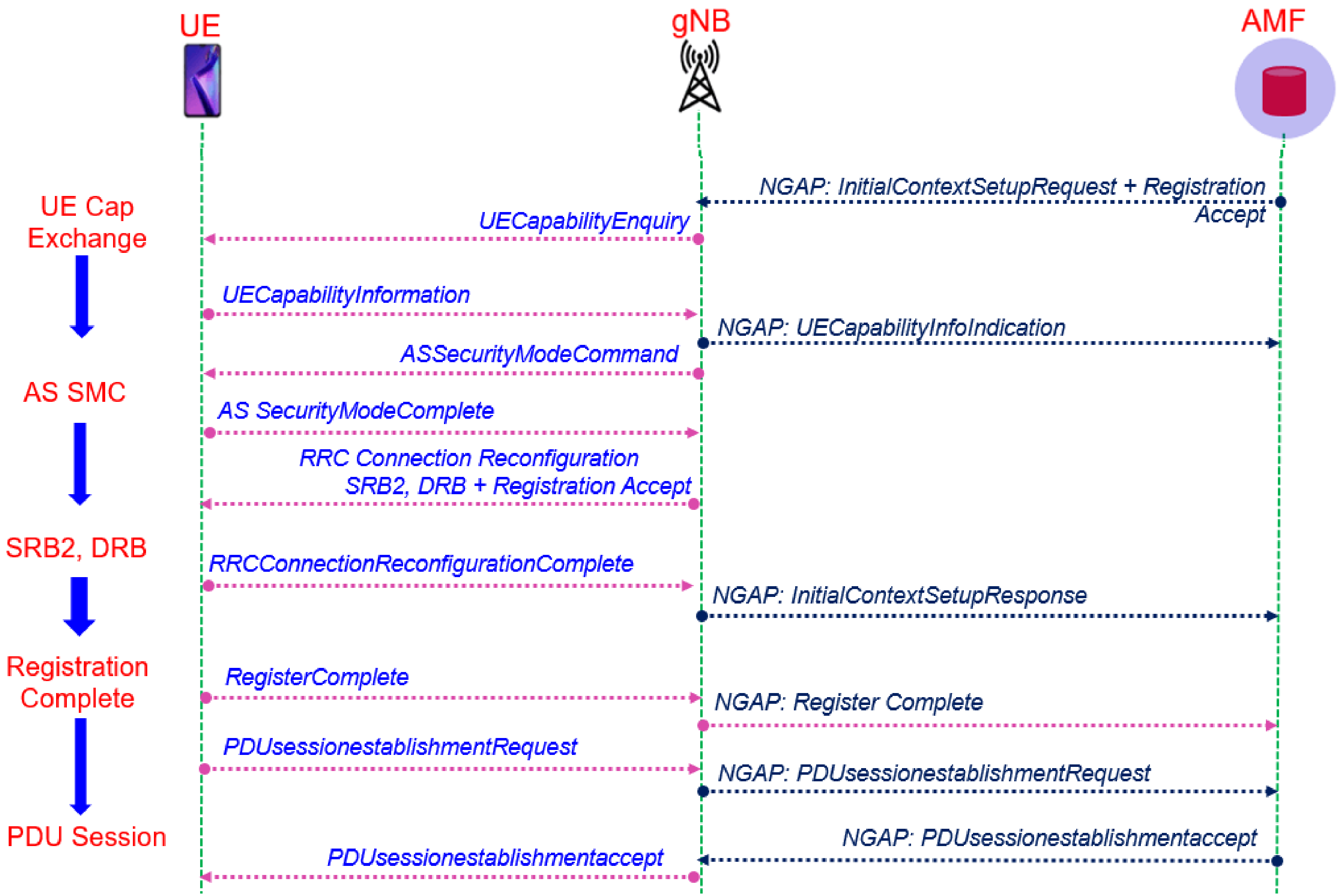
Ref: https://www.techplayon.com/5g-nr-sa-registration-attach-call-flow/

5G E2E Call flow -- SYNC to NAS SMC



Ref: https://www.techplayon.com/5g-nr-sa-registration-attach-call-flow/

5G E2E Call flow -- Registration to PDUsession



翔宇科技 EAGLETEK

Information for Quiz of 5G QoS lesson

- 5 questions for Acronyms of QoS parameters
- 5 questions for definitions/explanations of QoS parameters/QoS procedures
- Two questions for bonus

Thank You

https://www.peacedtogether.co.uk/blog/2018/07/thoughtfulthursday-29-thank-the-unappreciated/